The Geometry of Cochains on Sampled Vietoris-Rips Complexes (Private Working Draft)

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July 19, 2026

Abstract. We study the geometry of simplicial cochains on Vietoris–Rips complexes built from i.i.d. samples of a compact embedded manifold. By integrating differential forms over affine simplices, we define a cochain map from smooth forms to simplicial cochains and equip the latter with kernel-weighted inner products determined by ambient pairwise distances. At scales where the sampled complex recovers the homotopy type of the manifold, we prove quantitative high-probability convergence of these inner products and, under uniform sampling, of the associated codifferential energies to their continuum counterparts. As consequences, we obtain spectral upper bounds for the discrete Hodge Laplacian, convergence of harmonic representatives of discretized harmonic one-forms, and consistency of the harmonic smoothing step used to construct circular coordinates.

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1. Introduction

The Vietoris-Rips (VR) complex occupies a central role in topological data analysis (TDA) as it gives a concrete simplicial complex built from point-cloud (or finite metric) data at a scale parameter $\epsilon > 0$. Under manifold sampling assumptions, one has rigorous topological guarantees that the VR complex built from point samples is homotopy equivalent to the underlying topological space [18, 24, 30]. At the same time, Laplacian-based methods on graphs and simplicial complexes have become important tools for extracting geometric and harmonic information from data. This approach uses combinatorial Hodge theory, and requires an inner product structure on the simplicial cochain complex of the discrete space. Given any such choice of an inner product, Eckmann [15] developed a combinatorial Hodge Laplacian, whose kernel recovers cohomology, and furthermore yields a Hodge decomposition. While spectral properties of the combinatorial

Laplacian are studied in its own right [20, 29], these Laplacian-based methods have also been utilized in TDA, from defining circular coordinates using harmonic cochains [12] to more recent work on persistent Laplacians [41, 32, 10].

These constructions rely crucially on the choice of an inner product on the cochain complex; the standard choice is to declare the simplex-dual basis orthonormal. On a sampled manifold, however, simplex counts, Euclidean shapes, and sampling density all affect what a cochain norm should approximate. At present, there is limited theory justifying this choice, or more broadly explaining the relationship between the *geometry* of discrete simplicial cochains of VR complexes built from sampled data, and the corresponding continuum objects from the underlying space. In this article, we consider the setting of VR complexes built from a point cloud X sampled i.i.d. from an embedded compact manifold $M \subset \mathbb{R}^N$. Our main goal is to construct appropriate inner products and develop a de Rham-type discretization procedure from smooth differential k-forms to discrete simplicial cochains which is geometrically consistent with the L^2 Riemannian geometry of the underlying manifold. Our approach uses only the ambient Euclidean geometry of the sample, through affine simplices and pairwise distances, and does not require a priori knowledge of the intrinsic geometry of the manifold.

Our constructions are motivated by manifold learning methods used to understand various notions of convergence of graph Laplacians built from such sampled data to the Laplace-Beltrami operator on the underlying manifold. Early work [6, 11] built fully connected graphs with weights defined using Gaussian heat-kernels leading to qualitative spectral convergence results [5]. More recent approaches refined these results by considering a broader class of kernels on sparser ϵ-graphs (the 1-skeleton of the VR complex) and k-nearest neighbour graphs, and obtained quantitative convergence results [17, 9]. For $k = 0$, these results can be read as convergence of the L^2 geometry and Dirichlet energy of functions. Our aim is to extend this perspective to k-forms on the full VR cochain complex, focusing on the discretization direction: smooth forms are sent to cochains, but general cochains are not interpolated back to smooth forms. In particular, our work builds on the ϵ-graph approaches. The motivation is not only computational sparsity: these are also the finite-sample scales at which reconstruction theorems identify the VR complex with M.

Contributions and Outline. Let $X = (x_1, \dots, x_n) \subset \mathbb{R}^N$ be a finite collection of i.i.d. samples from a compact embedded manifold $M \subset \mathbb{R}^N$, and let X_ϵ denote the VR complex at scale $\epsilon > 0$. In Section 2, we begin by recalling preliminary geometric constructions used throughout the article, and discuss scale assumptions required for normal coordinate computations and distance comparisons used in our arguments.

Then in Section 3, we define our discretization of an ambient k-form $\alpha \in \Omega^k(\mathbb{R}^N)$ by integrating it over the affine Euclidean k-simplex spanned by each k-simplex of X_ϵ, writing the resulting simplicial cochain as $q\alpha \in C^k(X_\epsilon)$. In particular, this is a cochain map $q : \Omega^\bullet_c(\mathbb{R}^N) \to C^\bullet(X_\epsilon)$. We equip $C^k(X_\epsilon)$ with a kernel-weighted inner product $\langle \cdot, \cdot \rangle_n$ whose weights depend on the pairwise Euclidean distances among the vertices of a simplex. This induces a simplicial codifferential,

$$\hat{\delta} : C^k(X_\epsilon) \to C^{k-1}(X_\epsilon) \quad \text{as the adjoint of the simplicial coboundary} \quad \hat{d} : C^{k-1}(X_\epsilon) \to C^k(X_\epsilon).$$

This is our discrete analogue of the smooth codifferential,

$$\delta : \Omega^k(M) \to \Omega^{k-1}(M) \quad \text{as the adjoint of the exterior derivative} \quad d : \Omega^{k-1}(M) \to \Omega^k(M),$$

induced by the L^2 inner product $\langle \cdot, \cdot \rangle_M$ on $\Omega^\bullet(M)$. In Section 4, we present our main convergence results for the inner product and the codifferential. Let $U_\epsilon \subset \mathbb{R}^N$ be the ϵ-tubular neighbourhood of $M \subset \mathbb{R}^N$ with nearest point projection $\pi : U_\epsilon \to M$. For the codifferential convergence, our results are stated for *pullback forms* $\alpha = \pi^* \tilde{\alpha} \in \Omega^k(U_\epsilon)$, where $\tilde{\alpha} \in \Omega^k(M)$. At the relevant scales the affine simplices lie in U_ϵ, so this is sufficient to define $q\alpha$.

Theorem 1.1 (Informal). *For appropriate choices of ϵ_n, kernel weights, and normalizations, there exist constants $\sigma_k, \sigma_{\delta,k} > 0$ such that with high probability,*

(1) **(Topological Consistency)** *the VR complex X_{ϵ_n} is homotopy equivalent to M;*
(2) **(Inner Product Convergence)** *after the normalization and density correction specified below, for any $\alpha, \beta \in \Omega^k_c(\mathbb{R}^N)$, we have $\langle q\alpha, q\beta \rangle_n \to \sigma_k \langle \alpha, \beta \rangle_M$;*
(3) **(Codifferential Convergence)** *in the uniform-density pullback-form setting, for any $\tilde{\alpha}, \tilde{\beta} \in \Omega^k(M)$, we have $\langle \hat{\delta} q\alpha, \hat{\delta} q\beta \rangle_n \to \sigma_{\delta,k} \langle \delta\tilde{\alpha}, \delta\tilde{\beta} \rangle_M$, where $\alpha = \pi^*\tilde{\alpha}$ and $\beta = \pi^*\tilde{\beta}$.*

This result shows that we obtain convergence in probability of *geometric* structures within a bandwidth regime ϵ_n where X_{ϵ_n} has the correct homotopy type. The topological consistency result is stated formally in Proposition 4.8. The inner product and codifferential convergence results are stated formally in Theorem 4.1 and Theorem 4.5, where explicit rates are provided given a choice of ϵ_n. Furthermore, we highlight that these are *uniform* convergence results within the $\mathcal{C}^2$-ball of differential forms. The detailed proofs of the convergence results are deferred to Section 6 and Section 7.

Finally, in Section 5, we discuss some immediate consequences of our results related to the discrete Hodge Laplacian (normalized by a constant $\bar{\sigma}_{\delta,k}$ which arises from our convergence results),

$$\hat{\Delta}_{k,n} = \hat{\delta}_n \hat{d}_n + \bar{\sigma}_{\delta,k}^{-1} \hat{d}_n \hat{\delta}_n : C^k(X_\epsilon) \to C^k(X_\epsilon) \quad \text{with eigenvalues} \quad 0 \le \hat{\lambda}^k_{0,n} \le \hat{\lambda}^k_{1,n} \le \dots.$$

This is the discrete counterpart to the smooth Hodge Laplacian

$$\Delta_k = \delta d + d\delta : \Omega^k(M) \to \Omega^k(M) \quad \text{with eigenvalues} \quad 0 \le \lambda^k_0 \le \lambda^k_1 \le \dots.$$

For the following results, we fix a sequence ϵ_n such that our convergence results hold at the appropriate degrees. First, we provide a spectral upper bound, proved in Theorem 5.4.

Corollary 1.2. *For each fixed $j, k \in \mathbb{N}$, we have $\limsup_{n\to\infty} \hat{\lambda}^k_{j,n} \le \lambda^k_j$ in probability.*

In order to obtain stronger spectral convergence results, one must also develop *interpolation* methods to go from discrete cochains back to continuous differential forms. While this is beyond the scope of the present article, we view this as a promising avenue for future work. Despite this limitation, we consider the convergence of harmonic 1-forms by leveraging the known spectral convergence results for the graph Laplacian in [9]. This is proved in Proposition 5.8.

Corollary 1.3. *Let M be connected and $\omega \in \Omega^1(M)$ be a harmonic 1-form. For each n, let $\hat{\omega}_n \in C^1(X_{\epsilon_n})$ be the harmonic part of the pullback discretization $q\omega$ in the discrete Hodge decomposition. Then, $\|q\omega - \hat{\omega}_n\|_n \to 0$ in probability.*

As an application, Corollary 5.9 proves convergence of the smoothing step in the circular-coordinates method of [12], assuming the persistent-cohomology step has selected the integral class corresponding to the limiting harmonic form.

Related Work. Lerch and Wahl [28] define an empirical exterior calculus on certain alternating functions over a finite sample (called *empirical forms*) and prove high-probability bounds for the up-Laplacian energy of such forms. Their work is closely aligned with ours in treating differential forms directly at the level of sampled cochains; however their objects and results are different in several important ways. First, their weights are defined using the intrinsic heat kernel on the manifold, rather than from ambient distances between sample points. Second, their empirical forms are defined on all sample tuples, rather than on the simplices of a sampled VR complex, so the construction is not tied to the finite-scale topology guaranteed by reconstruction theorems. Finally, their main estimate is for the up-Laplacian energy and does not give convergence of the codifferential term needed for the full Hodge Laplacian on the sampled complex.

Other approaches avoid a finite simplicial cochain complex and instead approximate Hodge operators or form pairings directly. Spectral exterior calculus [7] builds forms from Laplace–Beltrami eigenfunctions and proves spectral convergence for Galerkin approximations of the 1-Hodge Laplacian. Diffusion-geometry methods [21] and neural point-forms [38] use carré-du-champ or Gram-field constructions to compute form inner products, Hodge energies, and learned form-comparison matrices from point clouds. Lê's empirical Hodge Laplacians [25] estimate tangent projections and curvature terms from a uniform point cloud, build empirical Hodge operators on projected ambient exterior powers, and prove spectral convergence together with recovery of the near-zero harmonic cluster. These methods have strong operator-theoretic and computational advantages, especially in avoiding large complexes and, in some cases, obtaining stronger spectral statements. The tradeoff is algebraic: they do not give a finite sampled complex in which cocycles, coboundaries, and chosen cohomology classes are available exactly at reconstruction scales; harmonic information is instead recovered from a near-zero spectral cluster on sampled vector data. In particular, they do not directly formulate convergence statements such as Corollary 1.3, where $\hat{\omega}_n$ is the harmonic representative of a specified cohomology class in the sampled VR complex.

A different line of work proves Hodge-theoretic convergence from controlled and deterministic intrinsic discretizations, such as covers, triangulations, or finite-element meshes. Mantuano [31] starts from an intrinsic ϵ-net, builds the nerve of the cover by geodesic balls, and proves a two-sided comparison between the smooth Hodge spectrum and the combinatorial spectrum on the resulting Čech cochains. Dodziuk's finite-difference approach and finite element exterior calculus [13, 4] use Whitney or finite-element forms, cochain projections, and mesh refinement to compare smooth and discrete Hodge theory on controlled complexes. Mantuano's result is especially close in spirit, since it also compares smooth and discrete Hodge theory through discretization and interpolation maps; however, these maps are built intrinsically from the Čech–de Rham double complex. The main distinction is that their discretizations are built from intrinsic geometric data or prescribed meshes; our complex is the sampled VR complex at various reconstruction scales, and its metric is defined only from ambient Euclidean distances.

AI Use Statement. The authors used ChatGPT and Claude as a research aid to explore proof strategies, refine hypotheses and constructions, and improve the exposition. The research direction and main results were developed by the authors, who independently verified all AI-assisted mathematical content and take full responsibility for the manuscript.

2. Geometric Preliminaries

In this article, we work with an m-dimensional compact Riemannian manifold M without boundary, which is smoothly embedded in Euclidean space, $\iota : M \hookrightarrow \mathbb{R}^N$. We equip M with the induced Riemannian structure g such that for $x \in M$ and $u, v \in T_xM$, we have

$$\langle u, v\rangle_x = g_x(u, v) = \langle \iota_* u, \iota_* v\rangle_{\mathbb{R}}, \tag{2.1}$$

where $\langle \cdot, \cdot \rangle_{\mathbb{R}}$ denotes the standard inner product on $\mathbb{R}^N$. We let vol_M denote the associated volume form.

2.1. Differential Forms and Hodge Operators. Let $\Omega^k(M)$ be the space of smooth differential k-forms on M; these are the sections of the vector bundle $\Lambda^k T^*M$. For each $x \in M$, the Riemannian metric (2.1) induces an inner product on T_x^*M, and furthermore on the exterior product $\Lambda^k T_x^* M$ defined by

$$\langle u_1 \wedge \ldots \wedge u_k, v_1 \wedge \ldots \wedge v_k\rangle_x = \det(\langle u_i, v_j\rangle_x).$$

Then, given two differential forms $\alpha, \beta \in \Omega^k(M)$, we can define the L^2 inner product by

$$\langle \alpha, \beta\rangle_M := \int_M \langle \alpha_x, \beta_x\rangle_x \, d\mathrm{vol}(x).$$

Note that we use $\langle \cdot, \cdot\rangle_x$ to denote all pointwise inner products and $\langle \cdot, \cdot \rangle_M$ to denote the L^2 inner product. We will also need to consider *weighted* L^2 inner products. Given a smooth function $f : M \to \mathbb{R}$, we define the f-weighted L^2 inner product by

$$\langle \alpha, \beta\rangle_M^f := \int_M \langle \alpha_x, \beta_x\rangle_x \, f(x) d\mathrm{vol}(x).$$

Next, let $d : \Omega^k(M) \to \Omega^{k+1}(M)$ be the exterior derivative such that $(\Omega^\bullet(M), d)$ is a cochain complex, $d^2 = 0$. The *codifferential* $\delta : \Omega^{k+1}(M) \to \Omega^k(M)$ is the formal adjoint to d, defined by

$$\langle d\alpha, \beta\rangle_M = \langle \alpha, \delta\beta\rangle_M, \quad \alpha \in \Omega^k(M), \quad \beta \in \Omega^{k+1}(M). \tag{2.2}$$

Finally, the *Hodge Laplacian* on $\Omega^k(M)$ is

$$\Delta_k = d\delta + \delta d : \Omega^k(M) \to \Omega^k(M), \tag{2.3}$$

and we say that $\omega \in \Omega^k(M)$ is a *harmonic k-form* if $\Delta_k \omega = 0$. Further details can be found in [22].

2.2. Ambient Forms and $\mathcal{C}^r$ Norms. While the differential forms in $\Omega^k(M)$ are the main limiting objects in this article, we wish to approximate them without prior knowledge of the manifold M. Thus, we primarily work with ambient differential forms. Let $\Omega_c^k(\mathbb{R}^N)$ denote the space of compactly-supported smooth k-forms on $\mathbb{R}^N$. Note that there is no loss of generality in considering compactly supported forms, as we primarily consider pullbacks to M, which is compact. Given ambient forms $\alpha, \beta \in \Omega_c^k(\mathbb{R}^N)$, we will often use

$$\langle \alpha, \beta\rangle_M := \langle \iota^*\alpha, \iota^*\beta\rangle_M$$

to denote the L^2 inner product of the pullback to M.

We will also use $\mathcal{C}^r$-norms to state uniform convergence results. Recall that an ambient k-form $\alpha \in \Omega^k(\mathbb{R}^N)$ can be expressed component-wise as $\alpha = \sum_{|I|=k} \alpha_I \, dz_I$ for multi-indices $I = (i_1 < \ldots < i_k)$. Then, on an open set $U \subset \mathbb{R}^N$, we define the $\mathcal{C}^r(U)$-norm

$$\|\alpha\|_{\mathcal{C}^r(U)} := \max_{|I|=k} \max_{|\mu| \le r} \sup_{z \in U} |\partial^\mu \alpha_I(z)|.$$

We will often use the ambient $\mathcal{C}^r$-unit ball

$$\mathcal{B}_{\mathbb{R}}^{k,r} := \{\alpha \in \Omega_c^k(\mathbb{R}^N) : \|\alpha\|_{\mathcal{C}^r(\mathbb{R}^N)} \le 1\}.$$

Throughout, $\bar{\nabla}$ denotes the standard flat connection on $\mathbb{R}^N$ used for ambient derivatives, while ∇ denotes the Levi-Civita connection on M used for intrinsic derivatives. For intrinsic forms $\omega \in \Omega^k(M)$, we define

$$\|\omega\|_{\mathcal{C}^r(M)} := \max_{0\leq j\leq r} \sup_{x\in M} |\nabla^j \omega_x|_x,$$

where the norm $|\cdot|_x$ is induced by the Riemannian metric g.

2.3. Reach and Tubular Neighbourhoods. We shall also use a canonical ambient extension of intrinsic forms to tubular neighbourhoods. First, we consider the notion of *reach*.

Definition 2.1. [1, Definition 2.1] For a closed subset $A \subset \mathbb{R}^N$, the *medial axis* $\mathrm{Med}(A)$ of A is the subset of $\mathbb{R}^N$ consisting of points which have at least two nearest neighbours,

$$\mathrm{Med}(A) = \{x \in \mathbb{R}^N \,:\, \exists p \neq q \in A,\, \|p - x\| = \|q - x\| = d_{\mathbb{R}}(x, A)\}.$$

The *reach of* A is defined as

$$\tau_A = d_{\mathbb{R}}(A, \mathrm{Med}(A)) = \inf_{x\in A,\, y\in \mathrm{Med}(A)} d_{\mathbb{R}}(x, y).$$

For $\epsilon > 0$, we define the *ϵ-tubular neighbourhood of M*, $M \subset U_\epsilon \subset \mathbb{R}^N$ by

$$U_\epsilon := \{z \in \mathbb{R}^N \,:\, d_{\mathbb{R}}(z, M) < \epsilon\}.$$

Then, if $\epsilon < \tau_M$, there exists a unique nearest point projection $\pi : U_\epsilon \to M$, which is smooth since M is smooth [27, Theorem 2]. For any $x \in M$, recall that we can decompose the ambient tangent space by $T_x\mathbb{R}^N = T_xM \oplus N_xM$, where $N_xM = (T_xM)^\perp$ is the orthogonal complement of the tangent space of M, called the *normal space*. In particular, we have $N_xM = \ker(d\pi_x)$.

2.4. Injectivity Radius and Normal Coordinates. In our analysis, we generally assume that we only have access to the ambient Euclidean geometry, while our approximation methods often rely on normal coordinates on the manifold. Here, we aim to obtain normal coordinates for open neighbourhoods defined by Euclidean balls. First, relate Euclidean balls and intrinsic balls. For any $x, y \in M$, we always have $d_{\mathbb{R}}(x, y) \leq d_M(x, y)$, so

$$B_{\epsilon,M}(x) \subset B_{\epsilon,\mathbb{R}}(x).$$

The other inclusion is controlled by the reach.

Lemma 2.2. [8, Lemma 3] *Let $\tau > 0$ be the reach of a compact manifold embedded in $\mathbb{R}^N$. For any $x, y \in M$ such that $d_{\mathbb{R}}(x, y) < 2\tau$, we have*

$$d_M(x, y) \leq 2\tau \arcsin\left(d_{\mathbb{R}}(x, y)/(2\tau)\right).$$

This implies that we have the chain of inclusions

$$B_\epsilon(M, x) \subset B_\epsilon(\mathbb{R}, x) \cap M \subset B_{S(\epsilon)}(M, x), \quad S(\epsilon) = 2\tau \arcsin(\epsilon/2\tau). \tag{2.4}$$

for any $\epsilon < 2\tau$. Next, we wish to ensure that the Riemannian exponential is well behaved.

Definition 2.3. Let M be a Riemannian manifold. We define the *injectivity radius* of M to be

$$\mathrm{inj}(M) := \sup\Big\{r > 0 \,:\, \exp_x : B_r(T_xM, 0) \to B_r(M, x) \text{ is a diffeomorphism for all } x \in M\Big\}.$$

We combine the above observations about reach and $\mathrm{inj}(M)$ and define a single constant,

$$\mathrm{inj}_{\mathbb{R}}(M) := \sup\{\epsilon \in (0, \min\{2\tau_M, 1\}) \,:\, S(\epsilon) < \mathrm{inj}(M)\}$$

such that we can work with a notion of normal coordinates across the entire manifold. We constrain $\mathrm{inj}_{\mathbb{R}}(M) \leq 1$ as a technical condition to simplify bounds. On a compact manifold, both the injectivity radius $\mathrm{inj}(M)$ [26, Lemma 6.16] and the reach τ_M [37, Prop. 14] are positive, which implies that $\mathrm{inj}_{\mathbb{R}}(M)$ is also positive.

Corollary 2.4. *Let $\epsilon < \mathrm{inj}_{\mathbb{R}}(M)$. Then for any $x \in M$, the restriction of the logarithm*

$$\log_x : B_\epsilon(\mathbb{R}^N, x) \cap M \to T_x M.$$

is a diffeomorphism onto its image.

Proof. Since $\epsilon < \mathrm{inj}_{\mathbb{R}}(M)$, then $S(\epsilon) < \mathrm{inj}(M)$ so that $\exp_x : B_{S(\epsilon)}(T_x M, 0) \to B_{S(\epsilon)}(M, x)$ is a diffeomorphism for all $x \in M$ with inverse $\log_x$. The inclusion $B_\epsilon(\mathbb{R}^N, x) \cap M \subset B_{S(\epsilon)}(M, x)$ from (2.4) shows that the restriction of $\log_x$ to $B_\epsilon(\mathbb{R}^N, x) \cap M$ is a diffeomorphism onto its image. □

We will often use this in the context of a change of variables to normal coordinates, and we denote the Jacobian of $\exp_x : B_\epsilon(T_x M, 0) \to M$ at $v \in B_\epsilon(T_x M, 0)$ by $J(x, v)$, and recall that

$$J(x,v) \leq 1 + C_M \|v\|^2. \tag{2.5}$$

To obtain comparisons between ambient and intrinsic displacements, we consider the ambient exponential map for $x \in M$,

$$\widetilde{\exp}_x = \iota \circ \exp_x : T_x M \to \mathbb{R}^N.$$

Recall the third order Taylor expansion of $\widetilde{\exp}_x$ as [33]

$$\widetilde{\exp}_x(v) = x + v + Q(v,v) + r(v), \quad v \in T_x M, \tag{2.6}$$

where Q is the second fundamental form, so $Q(v,v) \in N_x M$ is valued in the normal space. The remainder satisfies $|r(v)| \leq C_M \|v\|^3$. Then, if $\widetilde{\exp}_x(v) = y$, we have

$$\|v - (y - x)\| \lesssim \|v\|^2 \quad \text{and} \quad \big| \|v\|^2 - \|y - x\|^2 \big| \lesssim \|v\|^4, \tag{2.7}$$

where the latter inequality uses the fact that the cubic terms are trivial since $\langle v, Q(v,v) \rangle_{\mathbb{R}} = 0$.

Finally, we fix a global scale parameter for the remainder of the paper. The *convexity radius* is

$$\mathrm{cx}(M) = \sup\{r \in [0,\infty) \,:\, B_r(M,x) \text{ is strongly geodesically convex for all } x \in M\}. \tag{2.8}$$

Here, strongly geodesically convex means that any two points in the ball are joined by a unique minimizing geodesic contained in the ball. Because M is compact, $\mathrm{cx}(M) > 0$ [35, Section 6.4.2]. We fix a constant $\epsilon_0 = \epsilon_0(M) > 0$, which depends only on M such that

$$0 < 2\epsilon_0 < \mathrm{inj}_{\mathbb{R}}(M) \quad \text{and} \quad S(2\epsilon_0) < \mathrm{cx}(M).$$

All subsequent small-scale estimates will be stated for $0 < \epsilon < \epsilon_0$.

3. Discretization and Inner Products on VR Complex

We now introduce the discrete analogues of manifolds, differential forms, and inner products which were discussed in the previous section. The primary data that we work with are finite collections of point $X^{(n)} = (x_1, \ldots, x_n) \subset \mathbb{R}^N$ in $\mathbb{R}^N$, though for simplicity we will often refer to the point cloud as $X = X^{(n)}$. For $\epsilon > 0$, the *ϵ-Vietoris Rips (VR) complex* is a simplicial complex X_ϵ, with X as its vertex set, and $\sigma \subset X$ is a simplex if $\mathrm{diam}_{\mathbb{R}}(\sigma) < \epsilon$. In particular, the set of k-simplices of X_ϵ is given by

$$S^k_\epsilon(X) := \{\boldsymbol{x}_I = (x_{i_0}, \ldots, x_{i_k}) \subset X \,:\, I = (i_0 < \ldots < i_k),\, \mathrm{diam}_{\mathbb{R}}(\boldsymbol{x}_I) < \epsilon\}.$$

with tuples that differ by a permutation identified. Recall that the simplicial k-chains of X_ϵ is given by $C_k(X_\epsilon) := \mathbb{R}[S^k_\epsilon(X)]$, and the simplicial k-cochains $C^k(X_\epsilon)$ is the vector space of linear maps $a : C_k(X_\epsilon) \to \mathbb{R}$. By linearity, this is equivalent to specifying a map $a : S^k_\epsilon(X) \to \mathbb{R}$. Simplicial cochains $C^\bullet(X_\epsilon)$ are equipped with the usual differential

$$\hat{d} : C^k(X_\epsilon) \to C^{k+1}(X_\epsilon), \quad \hat{d}a(y_0, \ldots, y_{k+1}) = \sum_{i=0}^{k+1} (-1)^i a(y_0, \ldots, \hat{y}_i, \ldots, y_{k+1}).$$

In this article, we consider the VR complex of a point cloud sampled from an embedded manifold $M \subset \mathbb{R}^N$ such that $X = (x_1, \ldots, x_n) \subset M \subset \mathbb{R}^N$. We emphasize that we use the ambient metric from $\mathbb{R}^N$ to define X_ϵ. In this case, the simplicial complex X_ϵ acts as our discrete approximation of M, and simplicial cochains $C^k(X_\epsilon)$ act as the discrete analogue of differential forms.

3.1. Euclidean de Rham Map. Given an ambient differential form $\alpha \in \Omega^k(\mathbb{R}^N)$, and a point cloud $X \subset \mathbb{R}^N$, our aim is to discretize α as a simplicial cochain on X_ϵ. This discretization is performed via the classical *Euclidean de Rham map*, which we initially define as a function

$$q : \Omega^k_c(\mathbb{R}^N) \to C^\infty\left((\mathbb{R}^N)^{k+1}, \mathbb{R}\right) \quad \text{by} \quad q\alpha(\boldsymbol{y}) := \int_{\Delta(\boldsymbol{y})} \alpha,$$

for $\boldsymbol{y} = (y_0, \ldots, y_k) \in (\mathbb{R}^N)^{k+1}$, where $\Delta(\boldsymbol{y}) \subset \mathbb{R}^N$ is the Euclidean k-simplex defined as the convex hull of $\boldsymbol{y}$ in $\mathbb{R}^N$. We will also call q the *discretization map*.

Given a point cloud $X \subset \mathbb{R}^N$ and its associated VR complex X_ϵ, we restrict the de Rham map q to the k-simplices $S^k(X_\epsilon)$ in X_ϵ to obtain a map

$$q_{X,\epsilon} : \Omega^k_c(\mathbb{R}^N) \to C^k(X_\epsilon).$$

As the point cloud X and ϵ are usually clear from context, we will often suppress the subscripts and also denote this map by q. A key property of this discretization is that it is compatible with the smooth and discrete cochain structures.

Lemma 3.1. *For $X = (x_1, \ldots, x_n) \subset \mathbb{R}^N$ and $\epsilon > 0$, the discretization $q_{X,\epsilon}$ is a cochain map, $\hat{d}q = qd$.*

Proof. This is immediate by Stokes' theorem. □

A crucial property for our later approximations is that the discretization map is well approximated by evaluation against both ambient and intrinsic displacement vectors based at one of the vertices.

Lemma 3.2. *Let $\alpha \in \Omega^k(\mathbb{R}^N)$ and $\boldsymbol{y} = (y_0, \ldots, y_k) \in (\mathbb{R}^N)^{k+1}$. Let $\boldsymbol{w} = (y_1 - y_0, \ldots, y_k - y_0)$ be the Euclidean displacement vectors based at y_0. Then,*

$$q\alpha(\boldsymbol{y}) = \frac{\alpha_{y_0}(\boldsymbol{w})}{k!} + \frac{1}{(k+1)!} \sum_{i=1}^{k} (\bar{\nabla}_{w_i}\alpha)_{y_0}(\boldsymbol{w}) + R(\boldsymbol{w}), \quad |R| \lesssim \|\bar{\nabla}^2\alpha\|_\infty \|\boldsymbol{w}\|^{k+2}. \tag{3.1}$$

where $\bar{\nabla}$ is the usual covariant derivative on $\mathbb{R}^N$.

Proof. Let $\Delta^k = \{(u_1, \ldots, u_k) \in \mathbb{R}^k : u_i \geq 0, \sum_{i=1}^k u_i \leq 1\}$. We parametrize $\Delta(\boldsymbol{y})$ by

$$\sigma : \Delta^k \to \mathbb{R}^N, \quad \sigma(u_1, \ldots, u_k) = y_0 + \sum_{i=1}^k u_i w_i.$$

Using this parametrization, we obtain

$$q\alpha(\boldsymbol{y}) = \int_{\Delta(\boldsymbol{y})} \alpha = \int_{\Delta^k} \alpha_{\sigma(u)}(\boldsymbol{w}) du.$$

By fixing $\boldsymbol{w} = (w_1, \dots, w_k)$, and treating $\alpha_{\sigma(\cdot)}(\boldsymbol{w}) : \Delta^k \to \mathbb{R}$, the second order Taylor expansion at $u = 0$ is

$$\alpha_{\sigma(u)}(\boldsymbol{w}) = \alpha_{\sigma(0)}(\boldsymbol{w}) + \sum_i u_i (\bar{\nabla}_{w_i}\alpha)_{y_0}(\boldsymbol{w}) + R_2(u), \quad \text{where} \quad |R_2(u)| \lesssim \|\bar{\nabla}^2\alpha\|_\infty \|\boldsymbol{w}\|^{k+2}.$$

Then, integrating this expression, and using the fact that $\int_{\Delta^k} u_i du = \frac{1}{(k+1)!}$, we obtain (3.1). $\square$

Lemma 3.3. *Let $\alpha \in \Omega^k(\mathbb{R}^N)$ and $\boldsymbol{y} = (y_0, \dots, y_k) \in M^{k+1} \subset \mathbb{R}^N$ such that $\operatorname{diam}_{\mathbb{R}}(\boldsymbol{y}) < \operatorname{inj}_{\mathbb{R}}(M)$. Let $\boldsymbol{v} = (\log_{y_0}(y_1), \dots, \log_{y_0}(y_k)) \in T^k_{y_0}M$ be the intrinsic displacement vectors based at y_0. Then,*

$$q\alpha(\boldsymbol{y}) = \frac{\alpha_{y_0}(\boldsymbol{v})}{k!} + L(\boldsymbol{v}) + R(\boldsymbol{v}), \quad |R(\boldsymbol{v})| \lesssim \|\alpha\|_{C^2(\mathbb{R}^N)} \|\boldsymbol{v}\|^{k+2}$$

and $L : T^k_x M \to \mathbb{R}$ is homogeneous of degree $k+1$; in other words, $L(\lambda \boldsymbol{v}) = \lambda^{k+1} L(\boldsymbol{v})$ for all $\lambda \in \mathbb{R}$.

Proof. By Lemma 3.2 (and using the notation there), we obtain

$$q\alpha(\boldsymbol{y}) = \frac{\alpha_{y_0}(\boldsymbol{w})}{k!} + \frac{1}{(k+1)!} \sum_{i=1}^k (\bar{\nabla}_{w_i}\alpha)_{y_0}(\boldsymbol{w}) + R_0(\boldsymbol{y}). \tag{3.2}$$

By (2.7), we note that $|R_0(\boldsymbol{y})| \lesssim \|\bar{\nabla}^2\alpha\|_\infty \|\boldsymbol{v}\|^{k+2}$. Next, using the Taylor expansion for the exponential from (2.6), we get

$$w_i = \widetilde{\exp}_{y_0}(v_i) - y_0 = v_i + Q_{y_0}(v_i, v_i) + r_i, \quad \text{where} \quad \|r_i\| \lesssim \|v_i\|^3,$$

and Q_x is the second fundamental form of M. Applying this to the first term in (3.2), we obtain

$$\alpha_{y_0}(\boldsymbol{w}) = \alpha_{y_0}(\boldsymbol{v}) + \sum_{i=1}^k \alpha_{y_0}(v_1, \dots, Q_{y_0}(v_i, v_i), \dots, v_k) + E_1(\boldsymbol{v}),$$

where $|E_1(\boldsymbol{v})| \lesssim \|\alpha\|_\infty \|\boldsymbol{v}\|^{k+2}$. For the second term in (3.2), by linearity of $\bar{\nabla}_w$ in w, we get

$$\sum_{i=1}^k (\bar{\nabla}_{w_i}\alpha)_{y_0}(\boldsymbol{w}) = \sum_{i=1}^k (\bar{\nabla}_{v_i}\alpha)_{y_0}(\boldsymbol{v}) + E_2(\boldsymbol{v}),$$

where $|E_2| \lesssim \|\bar{\nabla}\alpha\|_\infty \|\boldsymbol{v}\|^{k+2}$. Thus, we set

$$L(\boldsymbol{v}) = \frac{1}{k!} \sum_{i=1}^k \alpha_{y_0}(v_1, \dots, Q_{y_0}(v_i, v_i), \dots, v_k) + \frac{1}{(k+1)!} \sum_{i=1}^k (\bar{\nabla}_{v_i}\alpha)_{y_0}(\boldsymbol{v}),$$

which is homogeneous of degree $k+1$, and the remainder is $R = R_0 + E_1 + E_2$. $\square$

We note that both of the above estimates can be applied with any chosen base vertex in $\boldsymbol{y} = (y_0, \dots, y_k)$ by reordering the vertices of the simplex; the only change is an orientation sign induced by the permutation.

3.2. Bilinear Forms and Inner Products. Next, we will consider a general class of inner products on $C^k(X_\epsilon)$, which we will show to be compatible with the manifold inner product. This is a higher-dimensional analogue of the kernel weights used in graph Laplacian convergence [11, 17, 9], where edges are assigned weights dependent on the distance between end points $\|x - y\|$. Here, we will define weights on k-simplices which depend on pairwise squared distances between vertices. We define the *pairwise squared distance function* as the map

$$p : (\mathbb{R}^N)^{k+1} \to [0, \infty)^{\binom{k+1}{2}} \quad \text{where} \quad p(\boldsymbol{y}) = (\|y_i - y_j\|^2)_{i<j},$$

and $\boldsymbol{y} = (y_0, \dots, y_k) \in (\mathbb{R}^N)^{k+1}$.

Definition 3.4. Let $k \in \mathbb{N}$. A family of maps $\kappa_\epsilon : (\mathbb{R}^N)^{k+1} \to \mathbb{R}$ for $\epsilon > 0$ is an *admissible VR kernel of degree k* if it has the form

$$\kappa_\epsilon(\boldsymbol{y}) = \Theta\left(\frac{p(\boldsymbol{y})}{\epsilon^2}\right) \quad \text{where} \quad \Theta : [0,\infty)^{\binom{k+1}{2}} \to [0,\infty)$$

is supported on the cube $[0,1]^{\binom{k+1}{2}} \subset [0,\infty)^{\binom{k+1}{2}}$, Lipschitz on $[0,1)^{\binom{k+1}{2}}$ (such that Θ may have discontinuities on the boundary of this cube) and $\Theta(p(\boldsymbol{y}))$ is invariant under the natural action of the permutation group Σ_{k+1} on $\boldsymbol{y}$,

$$\Theta(p(y_0, \ldots, y_k)) = \Theta(p(y_{\pi(0)}, \ldots, y_{\pi(k)})) \quad \text{for} \quad \pi \in \Sigma_{k+1}.$$

If $\Theta(\boldsymbol{z}) > 0$ for all $\boldsymbol{z} \in [0,1)^{\binom{k+1}{2}}$, then we say that κ_ϵ is an *strict VR kernel of degree k*.

In particular, by definition in terms of the pairwise squared distance function, Θ is invariant under the diagonal orthogonal action of $A \in O(\mathbb{R}^N)$,

$$\Theta(p(Ay_0, \ldots, Ay_k)) = \Theta(p(y_0, \ldots, y_k)).$$

One class of admissible VR kernels is given as follows, and will be used to study the codifferential. Let $\theta : [0,\infty) \to [0,\infty)$ be a Lipschitz function supported on $[0,1]$. We define

$$\kappa_{\theta,\epsilon} : (\mathbb{R}^N)^{k+1} \to \mathbb{R} \quad \text{by} \quad \kappa^k_{\theta,\epsilon}(\boldsymbol{z}) = \theta\left(\frac{s_k(\boldsymbol{z})}{\epsilon^2}\right) \quad \text{where} \quad s_k(\boldsymbol{z}) = \max_{0\leq i<j\leq k} \|z_i - z_j\|^2.$$

We use these admissible VR kernels and an additional weighting function to define our desired inner products on simplicial cochains.

Definition 3.5. Let $w : (\mathbb{R}^N)^{k+1} \to (c_w, \infty)$ be a symmetric smooth function bounded below by $c_w > 0$ and κ_ϵ be an admissible VR kernel. Let $\epsilon > 0$, $X = (x_1, \ldots, x_n) \subset \mathbb{R}^N$, and X_ϵ be the corresponding VR complex. We define the following family of bilinear forms, called *w-weighted kernel forms* on $C^k(X_\epsilon)$,

$$\langle a, b\rangle^{\kappa,w}_{n,\epsilon} := \binom{n}{k+1}^{-1} \frac{(k!)^2}{\epsilon^{k(m+2)}} \sum_{\boldsymbol{y}\in S^k_\epsilon(X)} a(\boldsymbol{y})b(\boldsymbol{y})\kappa_\epsilon(\boldsymbol{y})w(\boldsymbol{y}), \quad a, b \in C^k(X_\epsilon), \tag{3.3}$$

where $\boldsymbol{y} = (y_0, \ldots, y_k)$. If κ_ϵ is a strict VR kernel, then we call the above *w-weighted kernel inner products*.

Remark 3.6. Note that if κ_ϵ is only an admissible VR kernel, then there may exist simplices $\boldsymbol{y} \in S^k_\epsilon(X)$ such that $\kappa_\epsilon(\boldsymbol{y}) = 0$, and thus there exist cochains with trivial norm. Once κ_ϵ is a strict VR kernel, then this cannot happen. We note that our inner product results only require κ_ϵ to be admissible, and not necessarily strict, whereas our codifferential and discrete Laplacian results require strictness and genuine inner products.

We will often fix an admissible VR kernel and weight function, and suppress the superscripts by using $\langle\cdot,\cdot\rangle_{n,\epsilon}$ to simplify notation.

3.3. Probabilistic Inner Products as U-Statistics. Thus far, we have considered inner products on X_ϵ where $X \subset \mathbb{R}^N$ is a fixed finite point cloud. In our main results, we are concerned with the setting where $X = (x_1, \ldots, x_n)$ is a random collection of points which are i.i.d. sampled from a measure μ supported on M. We will assume that μ has a $\mathcal{C}^2$ density $\rho : M \to (0,\infty)$ with respect to the volume measure vol_M on M, which is bounded above and below,

$$\rho_- \leq \rho(x) \leq \rho_+ \tag{3.4}$$

for fixed constants $0 < \rho_- < \rho_+$. In particular, given two differential forms $\alpha, \beta \in \Omega^k(\mathbb{R}^N)$, we wish to understand the inner product $\langle q\alpha, q\beta \rangle_{n,\epsilon}^{\kappa,w}$ as a random variable as $n \to \infty$. In order to do this, we will view the inner product as a U-statistic.

Definition 3.7. Let $h : M^{k+1} \to \mathbb{R}$ be a symmetric function, which we call a *U-kernel*. Let $X = (x_1, \dots, x_n) \in M^n$ be i.i.d. samples from a probability measure μ on M. Then, the *U-statistic* corresponding to h is

$$\mathbb{U}_n[h](X) := \binom{n}{k+1}^{-1} \sum_{i_0 < \dots < i_k} h(x_{i_0}, \dots, x_{i_k}).$$

Fix a symmetric weight $w : M^{k+1} \to \mathbb{R}$ and an admissible VR kernel κ_ϵ. Given $\alpha, \beta \in \Omega^k(\mathbb{R}^N)$, we define the U-kernel $h_\epsilon(\alpha, \beta) : M^{k+1} \to \mathbb{R}$ by

$$h_\epsilon(\alpha, \beta)(\boldsymbol{y}) := \frac{(k!)^2}{\epsilon^{k(m+2)}} q\alpha(\boldsymbol{y}) q\beta(\boldsymbol{y}) \kappa_\epsilon(\boldsymbol{y}) w(\boldsymbol{y}). \tag{3.5}$$

Then, by definition of the U-statistic and the inner product in (3.3), we have

$$\langle q\alpha, q\beta \rangle_{n,\epsilon}^{\kappa,w} = \mathbb{U}_n[h_\epsilon(\alpha, \beta)](X).$$

To understand the limit as $n \to \infty$, we consider the following strong law of large numbers for U-statistics.

Theorem 3.8. [36, Section 5.4, Theorem A] *Let (Z, μ) be a probability space, and let $h : Z^{k+1} \to \mathbb{R}$ be a symmetric measurable function such that $\mathbb{E}_{\mu^{k+1}}[|h|] < \infty$. Let $X = (x_1, \dots, x_n) \in Z^n$ be a collection of i. i. d. samples from μ. Then, for all $n \geq k+1$, almost surely*

$$\lim_{n \to \infty} \mathbb{U}_n[h] = \mathbb{E}_{\mu^{k+1}}[h] = \mathbb{E}_{\mu^n}[\mathbb{U}_n[h]]$$

As an example, let $\alpha, \beta \in \Omega^k(\mathbb{R}^N)$. We define the $(k+1, \epsilon)$*-fat diagonal of M* with respect to the ambient metric by

$$D_\epsilon^k(M) := \{\boldsymbol{y} \in M^{k+1} : \operatorname{diam}_{\mathbb{R}}(\boldsymbol{y}) < \epsilon\}. \tag{3.6}$$

Then, for a fixed $\epsilon > 0$, almost surely,

$$\lim_{n \to \infty} \langle q\alpha, q\beta \rangle_{n,\epsilon}^{w} = \frac{(k!)^2}{\epsilon^{k(m+2)}} \int_{D_\epsilon^k(M)} q\alpha(\boldsymbol{y}) q\beta(\boldsymbol{y}) w(\boldsymbol{y}) \kappa_\epsilon(\boldsymbol{y}) \, d\mu^{k+1}(\boldsymbol{y}) =: \mathbb{E}_{\mu^{k+1}}[h_\epsilon].$$

4. Uniform Convergence of Inner Products and Codifferential

Having constructed kernel-weighted inner products on cochains of sampled VR complexes, we will now state our main convergence results, along with immediate corollaries. We postpone the proof of the two main results to Section 6 and Section 7.

4.1. Uniform Convergence of Inner Products.

We now show that these discrete inner products recover the continuum L^2 geometry of differential forms on M. The main result of this section is a uniform convergence theorem for the empirical inner products of discretized forms.

Theorem 4.1. *Let $M \subset \mathbb{R}^N$ be an embedded manifold of dimension m and μ be a probability measure on M with C^2 density ρ satisfying (3.4) with respect to the volume measure. Let $w : M^{k+1} \to \mathbb{R}$ be a smooth symmetric function, and κ_ϵ be an admissible VR kernel whose defining function Θ is nonzero on a subset* *of* $[0,1)^{\binom{k+1}{2}}$ *of positive measure. Let*

$$r_{k,m} = \min\left\{\frac{1}{m+2}, \frac{2}{km+2m+2}\right\} \quad \textit{and} \quad \epsilon_n \asymp n^{-1/(km+2m+2)}.$$

Let $p \in (0,1)$*. With probability at least* $1-p$*, for* n *such that*

$$\epsilon_n < \epsilon_0 \quad \textit{and} \quad \log(8/p) \leq n^{(2m+2)/(km+2m+2)},$$

we have

$$\sup_{\alpha,\beta \in \Omega_c^k(\mathbb{R}^N), \alpha,\beta \neq 0} \frac{|\langle q\alpha, q\beta\rangle_{n,\epsilon_n}^{\kappa,w} - \sigma_k^{\Theta}\langle \alpha,\beta\rangle_M^{P_\Delta}|}{\|\alpha\|_{\mathcal{C}^2(\mathbb{R}^N)}\|\beta\|_{\mathcal{C}^2(\mathbb{R}^N)}} \lesssim n^{-r_{k,m}} + \sqrt{\frac{\log(1/p)}{n}} + \frac{\log(1/p)}{n^{(2m+2)/(km+2m+2)}},$$

where $P_\Delta(x) = w(x,\ldots,x)\rho(x)^{k+1}$ *and* $\sigma_k^{\Theta} > 0$ *is a fixed constant.*

Remark 4.2. In Theorem 4.1, the limiting weight is $P_\Delta(x) = w(x,\ldots,x)\rho(x)^{k+1}$. Thus, choosing

$$w(y_0,\ldots,y_k) = \prod_{i=0}^{k}\rho(y_i)^{-1}$$

cancels the sampling density and recovers the usual L^2 inner product on M, up to the fixed kernel constant σ_k^{Θ}. In applications, ρ is typically unknown, so one should instead use a plug-in weight built from a kernel density estimate $\hat{\rho}$ of the sampling density; standard uniform consistency results for KDE on manifolds then control the additional error [11].

We can obtain a simpler statement for a large range of k and m for fixed probability p.

Corollary 4.3. *With the same hypotheses as Theorem 4.1. Suppose* $km \geq 2$*, then for a fixed* $p \in (0,1)$ *and sufficiently large* n*, we have with probability at least* $1-p$*,*

$$\sup_{\alpha,\beta \in \Omega_c^k(\mathbb{R}^N), \alpha,\beta \neq 0} \frac{|\langle q\alpha, q\beta\rangle_{n,\epsilon_n}^{\kappa,w} - \sigma_k^{\Theta}\langle \alpha,\beta\rangle_M^{P_\Delta}|}{\|\alpha\|_{\mathcal{C}^2(\mathbb{R}^N)}\|\beta\|_{\mathcal{C}^2(\mathbb{R}^N)}} \lesssim_p n^{-2/(km+2m+2)}.$$

We prove the theorem in several steps. First, by bilinearity of both inner products, it suffices to establish the estimate uniformly over the $\mathcal{C}^2$-unit ball

$$\mathcal{B}_{\mathbb{R}} := \{\alpha \in \Omega_c^k(\mathbb{R}^N) : \|\alpha\|_{\mathcal{C}^2(\mathbb{R}^N)} \leq 1\}. \tag{4.1}$$

Indeed, the general normalized estimate follows by applying the unit-ball estimate to $\alpha/\|\alpha\|_{\mathcal{C}^2(\mathbb{R}^N)}$ and $\beta/\|\beta\|_{\mathcal{C}^2(\mathbb{R}^N)}$. Next, for $\alpha,\beta \in \mathcal{B}_{\mathbb{R}}$, we decompose the error,

$$\left|\langle q\alpha, q\beta\rangle_{n,\epsilon_n}^{\kappa,w} - \sigma_k^{\Theta}\langle\alpha,\beta\rangle_M^{P_\Delta}\right| \leq \left|\langle q\alpha, q\beta\rangle_{n,\epsilon_n}^{\kappa,w} - \mathbb{E}\langle q\alpha, q\beta\rangle_{n,\epsilon}^{\kappa,w}\right| + \left|\mathbb{E}\langle q\alpha, q\beta\rangle_{n,\epsilon}^{\kappa,w} - \sigma_k^{\Theta}\langle\alpha,\beta\rangle_M^{P_\Delta}\right|, \tag{4.2}$$

where the first term is a probabilistic variance term and the second term is a deterministic bias term.

4.2. Codifferential Convergence. To relate the continuous Hodge Laplacian on $\Omega^k(M)$ and the discrete Hodge Laplacian on $C^k(X_\epsilon)$, we study the relationship between the smooth and discrete codifferentials

$$\delta : \Omega^k(M) \to \Omega^{k-1}(M) \quad \text{and} \quad \hat{\delta} : C^k(X_\epsilon) \to C^{k-1}(X_\epsilon),$$

which are defined as the adjoints of the exterior derivative $d : \Omega^{k-1}(M) \to \Omega^k(M)$ (2.2) and the simplicial coboundary map $\hat{d} : C^{k-1}(X_\epsilon) \to C^k(X_\epsilon)$ with respect to the corresponding smooth and discrete inner products. Note that we exclusively work with *strict VR kernels* such that the resulting bilinear form is indeed an inner product (see Remark 3.6). Furthermore, we work in the unweighted uniform-density case, $w \equiv 1$ and $\rho \equiv \rho_0 = \mathrm{vol}(M)^{-1}$, in order to keep the notation focused on the codifferential argument.

We also restrict our attention to forms $\alpha \in \Omega^k(U_\epsilon)$ on a tubular neighbourhood U_ϵ such that $\alpha = \pi^*\tilde{\alpha}$ for some $\tilde{\alpha} \in \Omega^k(M)$; we refer to such forms as *pullback forms*. Along M, we identify α with $\tilde{\alpha}$ and write $\delta\alpha$ for the intrinsic codifferential of $\tilde{\alpha}$. Since $\epsilon < \epsilon_0$, the Euclidean simplices considered below lie in the tubular neighbourhood where these pullback forms are defined. Finally, we work with a certain class of strict VR kernels which are bounded below on the open Vietoris–Rips scale.

Definition 4.4. Let $\theta : [0,\infty) \to [0,\infty)$ be a non-increasing function supported on $[0,1]$, Lipschitz on $[0,1)$, satisfying $\theta(1) = 0$, and $c_\theta = \inf_{t\in[0,1)} \theta(t) > 0$. For $\boldsymbol{z} = (z_0,\dots,z_r) \in (\mathbb{R}^N)^{r+1}$, define

$$\kappa^r_{\theta,\epsilon}(\boldsymbol{z}) = \theta\left(\frac{s_r(\boldsymbol{z})}{\epsilon^2}\right), \qquad s_r(\boldsymbol{z}) = \max_{0\le i<j\le r} \|z_i - z_j\|^2.$$

For each r, the kernels $\kappa^r_{\theta,\epsilon}$ are strict VR kernels because θ is bounded below on $[0,1)$. The monotonicity of θ is not used in the local codifferential expansion below, but is retained for compatibility with the graph-Laplacian comparison in Section 5.

The main result is the following uniform convergence for the codifferential.

Theorem 4.5. *Let $M \subset \mathbb{R}^N$ be an embedded manifold of dimension m, let $1 \le k \le m$, and assume that the sampling density is uniform, $\rho \equiv \rho_0 = \mathrm{vol}(M)^{-1}$. Let θ satisfy Definition 4.4, and let*

$$r^\delta_{k,m} = \frac{2}{2mk+5m+6} \quad \text{and} \quad \epsilon_n \asymp n^{-r^\delta_{k,m}}.$$

Set $\sigma^\theta_{\delta,k} = k^{(m+4)/2}\sigma_\Lambda$, where σ_Λ is the deterministic moment constant identified in Proposition 7.3. Let $p \in (0,1)$. With probability at least $1-p$, for n such that

$$\epsilon_n < \epsilon_0 \quad \text{and} \quad \log(12/p) \le n^{(3m+6)/(2mk+5m+6)},$$

we have, where we use $\alpha = \pi^*\tilde{\alpha}$ and $\beta = \pi^*\tilde{\beta}$ to denote pullback forms, and $\kappa = \kappa^{k-1}_{\theta,\epsilon_n}$,

$$\sup_{\substack{\tilde{\alpha},\tilde{\beta}\in\Omega^k(M)\\ \tilde{\alpha},\tilde{\beta}\neq 0}} \frac{\left|\langle \hat{\delta} q\alpha, \hat{\delta} q\beta\rangle^\kappa_{n,\epsilon_n} - \rho_0^{k+2}\sigma^\theta_{\delta,k}\langle \delta\tilde{\alpha}, \delta\tilde{\beta}\rangle_M\right|}{\|\tilde{\alpha}\|_{C^2(M)}\|\tilde{\beta}\|_{C^2(M)}} \lesssim n^{-r^\delta_{k,m}} + \frac{\sqrt{\log(1/p)}}{n^{1/2-r^\delta_{k,m}}} + \frac{\log(1/p)}{n^{(3m+2)/(2mk+5m+6)}}.$$

The implicit constants in this codifferential estimate may depend on M, m, k, the fixed kernel θ, and the uniform density ρ_0, but not on n, p, or the forms.

4.3. Topology of the VR Complex. The convergence results above provide analytic statements about the geometric properties of cochains on the sampled VR complex. A key advantage of our discretization framework is the realization of these discretized forms as cochains on an underlying simplicial complex. In particular, we show in this section that the scale parameters ϵ_n required for convergence result in VR complexes X_{ϵ_n} which are homotopy equivalent to M; and furthermore, that the discretization map q restricted to M yields a quasi-isomorphism.

There have been a series of works which have studied the topological properties of VR complexes, beginning with Hausmann [18]. For $\epsilon > 0$, let M_ϵ denote the *intrinsic* Vietoris-Rips complex of M, a simplicial complex with M as its vertex set and $\sigma \subset M$ with $|\sigma| = k+1$ is a k-simplex if $\mathrm{diam}_M(\sigma) < \epsilon$. By placing a total order on the elements of M, Hausmann defined a map $T : |M_\epsilon| \to M$, where $|M_\epsilon|$ is the geometric realization of M_ϵ as follows. On an ordered simplex $\sigma = [p_0, \dots, p_k]$, the map is defined inductively as the geodesic cone from the first vertex p_0. If $z = \sum_{i=0}^k t_i p_i$ and $t_0 < 1$, set $z' = (1-t_0)^{-1} \sum_{i=1}^k t_i p_i$ and let $T(z) = \gamma(1-t_0)$, where γ is the minimizing geodesic from p_0 to $T(z')$; if $t_0 = 1$, set $T(z) = p_0$. We consider the statement of Hausmann's theorem from [30, Theorem 4], which uses the convexity radius from (2.8).

Theorem 4.6. [18] *For any $0 < \epsilon < \mathrm{cx}(M)$, the map T is a homotopy equivalence.*

This result provides homotopy equivalence for the intrinsic VR complex. Later on, [24] provided an analogous result for VR complexes built from a finite sample X with small Hausdorff distance from M, and the specific scale parameters required were quantified in [30]. For our results, we consider the ambient affine realization $A : |X_\epsilon| \to \mathbb{R}^N$ of our sampled VR complex. We note that if $\epsilon < \epsilon_0$, then $A(|X_\epsilon|) \subset U_\epsilon$, where U_ϵ is the tubular neighbourhood of M. Define

$$F = \pi \circ A : |X_\epsilon| \to M,$$

and we will adapt the result from [30] to our setting.

Lemma 4.7. *Assume that M is connected. Fix $0 < \zeta < 1/14$. Suppose $h = d_H^{\mathbb{R}}(M, X) < \zeta\epsilon$, where $d_H^{\mathbb{R}}$ is the Hausdorff distance in $\mathbb{R}^N$ and $\epsilon < \epsilon_0$. Furthermore, suppose $\epsilon \leq A_\zeta \tau_M$, where $A_\zeta > 0$ is an explicit constant given in* [30, Theorem 18]. *Then $F : |X_\epsilon| \to M$ is a homotopy equivalence.*

Proof. The Gromov-Hausdorff condition induces a (possibly non-continuous and non-unique) vertex map $\phi : M \to X$ satisfying $\|p - \phi(p)\| < \zeta\epsilon$. This extends to a simplicial map

$$\Phi : M_{(1-2\zeta)\epsilon} \to X_\epsilon,$$

whose realization $|\Phi|$ is a homotopy equivalence [30, Theorem 18] and where $M_{(1-2\zeta)\epsilon}$ is the VR complex of M at scale $(1-2\zeta)\epsilon$. Let $T : M_{(1-2\zeta)\epsilon} \to M$ be Hausmann's geodesic realization. Our aim is to show that $F \circ |\Phi| \simeq T$.

Let $\sigma = [p_0, \dots, p_k]$ be a simplex of $M_{(1-2\zeta)\epsilon}$. The map T sends $|\sigma|$ into $B_{(1-2\zeta)\epsilon}(M, p_0) \subset B_{S(2\epsilon)}(M, p_0)$. For $z \in |\sigma|$, $A(|\Phi|(z))$ lies in the convex hull of the points $\phi(p_i)$, and

$$\|\phi(p_i) - p_0\| \leq \|\phi(p_i) - p_i\| + d_M(p_i, p_0) < (1-\zeta)\epsilon.$$

Hence $A(|\Phi|(z)) \in B_{(1-\zeta)\epsilon}(\mathbb{R}^N, p_0)$. Because the projection moves $A(|\Phi|(z))$ by less than ϵ, we have $\|F(|\Phi|(z)) - p_0\| < 2\epsilon$. By (2.4), $F(|\Phi|(z)) \in B_{S(2\epsilon)}(M, p_0)$. As we assume $S(2\epsilon) < \mathrm{cx}(M)$, this ball is strongly geodesically convex. The unique minimizing geodesics from $F(|\Phi|(z))$ to $T(z)$ therefore define simplexwise homotopies which agree on faces, giving $F \circ |\Phi| \simeq T$. Finally, since $|\Phi|$ and T are homotopy equivalences, so is F. □

Now we can show that F is a homotopy equivalence with high probability within the convergence regimes of Theorem 4.1 and Theorem 4.5.

Proposition 4.8. *Assume that M is connected. Let $X^{(n)} = (x_1, \dots, x_n) \subset M$ be sampled i.i.d. from a density satisfying* (3.4). *Let $\epsilon_n \to 0$ be deterministic and suppose that*

$$\frac{n\epsilon_n^m}{\log(1/\epsilon_n)} \to \infty. \tag{4.3}$$

Let $A_n : |X^{(n)}_{\epsilon_n}| \to \mathbb{R}^N$ be the affine realization map, and set $F_n = \pi \circ A_n : |X^{(n)}_{\epsilon_n}| \to M$. Then,

$$\lim_{n\to\infty} \mathbb{P}\left[F_n \textit{ is a homotopy equivalence}\right] = 1$$

On this event, $|X^{(n)}_{\epsilon_n}| \simeq M$ and the following pullback discretization is a quasi-isomorphism,

$$q_n^\pi : \Omega^\bullet(M) \to C^\bullet(X^{(n)}_{\epsilon_n}), \qquad (q_n^\pi \omega)(\sigma) = \int_{\Delta(\sigma)} \pi^* \omega.$$

Proof. Fix $0 < \zeta < 1/14$. There exists a constant $c_M > 0$ such that $\mathrm{vol}_M(B_r(M,x)) \geq c_M r^m$ for all $x \in M$ and $0 < r < \mathrm{inj}(M)$. Thus, for $0 < r < \mathrm{inj}(M)$, take an intrinsic $r/2$-net $z_1, \dots, z_L$ with $L \leq C_M r^{-m}$. The lower density bound gives $\mu(B_{r/2}(M,z_i)) \geq \rho_- c_M (r/2)^m$. If every such ball contains a sample point, then for every $p \in M$ there are z_i and $x_j \in X^{(n)}$ with $d_M(p,z_i) < r/2$ and $d_M(z_i,x_j) < r/2$. Since $d_{\mathbb{R}} \leq d_M$ on M, this implies $d_H^{\mathbb{R}}(M, X^{(n)}) < r$. By independence and the lower density bound, we have

$$\mathbb{P}\left[X^{(n)} \cap B_{r/2}(M,z_i) = \emptyset\right] = (1 - \mu(B_{r/2}(M,z_i)))^n \leq \exp(-n\mu(B_{r/2}(M,z_i))) \leq \exp(-c_M n r^m).$$

Then, by a union bound, we obtain

$$\mathbb{P}\left[d_H^{\mathbb{R}}(M, X^{(n)}) \geq r\right] \leq \sum_{i=1}^{L} \mathbb{P}\left[X^{(n)} \cap B_{r/2}(M,z_i) = \emptyset\right] \leq C_M r^{-m} \exp(-c_M n r^m). \tag{4.4}$$

Set $r = \zeta\epsilon_n$, and since $\epsilon_n \to 0$, for all sufficiently large n we have $\epsilon_n, r < \epsilon_0$, and $\epsilon_n \leq A_\zeta \tau_M$. On the complementary event, the final hypothesis of Lemma 4.7 that $d_H^{\mathbb{R}}(M, X^{(n)}) < \zeta\epsilon_n$ holds, so F_n is a homotopy equivalence. Then, (4.3) implies that the probability in (4.4) converges to 0.

For the quasi-isomorphism, let σ be an oriented simplex of $X^{(n)}_{\epsilon_n}$. The affine map A_n identifies $|\sigma|$ with the Euclidean simplex $\Delta(\sigma)$, and therefore

$$(q_n^\pi \omega)(\sigma) = \int_{\Delta(\sigma)} \pi^*\omega = \int_{|\sigma|} F_n^* \omega.$$

Thus q_n^π is the simplicial de Rham integration map applied to $F_n^*\omega$. Because F_n is a homotopy equivalence, F_n^* is a quasi-isomorphism; and since simplicial de Rham integration is a quasi-isomorphism by the simplicial de Rham theorem [42], so is q_n^π. ☐

Finally, Proposition 4.8 applies at the bandwidth scales used in the analytic convergence theorems. If $\epsilon_n \asymp n^{-a}$, then (4.3) is equivalent to $am < 1$. The choices in Theorem 4.1 and Theorem 4.5 have $a = 1/(km + 2m + 2)$ and $a = 2/(2mk + 5m + 6)$, respectively, and both satisfy $am < 1$. Therefore, when M is connected, for any fixed $p \in (0,1)$, the corresponding analytic estimate and the homotopy-equivalence/quasi-isomorphism conclusion above hold simultaneously with high probability.

5. The Discrete Laplacian

The discrete Laplacian on the VR complex is the spectral object attached to the cochain geometry constructed in the previous sections. The estimates in Theorems 4.1 and 4.5 give high-probability error bounds for the normalized discrete inner products and codifferential pairings after the prescribed choice of bandwidth. We record the resulting quantitative consequences for eigenvalue bounds, harmonic representatives, and circular coordinates.

Throughout this section we work in the unweighted uniform setting of the main codifferential Theorem 4.5: $w \equiv 1$ and $\rho \equiv \rho_0 = \mathrm{vol}(M)^{-1}$. We denote the point cloud by $X^{(n)} = (x_1, \dots, x_n) \subset M$. When $\alpha \in \Omega^k(M)$ is an intrinsic form, we write $q\alpha := q_n^\pi \alpha$ to simplify notation. We fix θ and kernels $\kappa_{\theta,\epsilon}^k$ satisfying Definition 4.4, but consider *normalized inner products*

$$\langle a, b\rangle_{n,\epsilon} := \frac{1}{\rho_0^{k+1}\sigma_k^\theta}\binom{n}{k+1}^{-1}\frac{(k!)^2}{\epsilon^{k(m+2)}}\sum_{\boldsymbol{y}\in S_\epsilon^k(X^{(n)})} a(\boldsymbol{y})b(\boldsymbol{y})\kappa_{\theta,\epsilon}^k(\boldsymbol{y}), \quad a, b \in C^k(X_\epsilon^{(n)}),$$

where $\sigma_k^\theta > 0$ is the constant from Theorem 4.1, and we take $\sigma_0^\theta = 1$. However, the constant $\sigma_{\delta,k}^\theta > 0$ from the codifferential convergence in Theorem 4.5 depends on normalizations in adjacent degrees, so we have the residual constant

$$\bar{\sigma}_{\delta,k} := \frac{\sigma_{k-1}^\theta}{(\sigma_k^\theta)^2}\sigma_{\delta,k}^\theta,$$

such that the normalized codifferential pairing converges to $\bar{\sigma}_{\delta,k}\langle\delta\alpha,\delta\beta\rangle_M$. In this section, we will also need the inner product in dimension zero. We define this as

$$\langle f, g\rangle_{n,\epsilon} = \frac{1}{\rho_0 n}\sum f(x_i)g(x_i).$$

for $f, g \in C^0(X_{\epsilon_n})$.

Remark 5.1. While the results in this section are stated as convergence in probability statements, the arguments are entirely quantitative. The high-probability estimates in Theorems 4.1 and 4.5 were stated with bandwidths chosen to give explicit rates, but the proofs give convergence throughout a range of bandwidths. When several estimates are used simultaneously, we choose one bandwidth in the common range and use a union bound. For convergence of the degree-k Laplacian, we use a bandwidth

$$\epsilon_n = n^{-a} \quad \text{satisfying} \quad 0 < a < \frac{1}{m(k+2)+2}, \tag{5.1}$$

which provides inner-product convergence in degrees k and $k+1$ and, when $k \geq 1$, the degree-k codifferential convergence. Quantitative versions are obtained by inserting the corresponding error bounds into the arguments below. We suppress these inherited expressions as they would obscure the simple form of the consequences.

5.1. Laplacians and Rayleigh Quotients.

Recall the smooth Hodge Laplacian from (2.3)

$$\Delta_k = d\delta + \delta d \quad \text{where} \quad 0 \leq \lambda_0^{(k)} \leq \lambda_1^{(k)} \leq \cdots$$

denotes the eigenvalues of Δ_k, repeated with multiplicity and listed in nondecreasing order. The *smooth Hodge energy* and the corresponding smooth Rayleigh quotient are defined by

$$Q_k(\alpha) := \langle\Delta_k\alpha, \alpha\rangle_M = \|d\alpha\|_M^2 + \|\delta\alpha\|_M^2 \quad \text{and} \quad R_k(\alpha) := \frac{Q_k(\alpha)}{\|\alpha\|_M^2} \quad \text{for} \quad \alpha \neq 0,$$

where the δ term in Q_k is omitted for $k = 0$. By the usual Courant–Fisher characterization,

$$\lambda_j^{(k)} = \min_{\substack{V\subset\Omega^k(M)\\ \dim V = j+1}} \max_{0\neq\alpha\in V} R_k(\alpha). \tag{5.2}$$

Next, we define the corresponding discrete objects. Let

$$\hat{d}_n : C^\bullet(X_{\epsilon_n}^{(n)}) \to C^{\bullet+1}(X_{\epsilon_n}^{(n)}) \quad \text{and} \quad \hat{\delta}_n : C^\bullet(X_{\epsilon_n}^{(n)}) \to C^{\bullet-1}(X_{\epsilon_n}^{(n)})$$

denote the simplicial coboundary and its adjoint. We define the *normalized Rips Laplacian* by

$$\widehat{\Delta}_{k,n} = \hat{\delta}_n \hat{d}_n + \bar{\sigma}_{\delta,k}^{-1} \hat{d}_n \hat{\delta}_n \quad \text{where} \quad 0 \leq \widehat{\lambda}_{0,n}^{(k)} \leq \widehat{\lambda}_{1,n}^{(k)} \leq \cdots \tag{5.3}$$

are the eigenvalues of $\widehat{\Delta}_{k,n}$, again repeated with multiplicity. The second term is omitted when $k = 0$.

Remark 5.2. Let $\Delta_{k,n}^{\mathrm{std}} := \hat{\delta}_n \hat{d}_n + \hat{d}_n \hat{\delta}_n$ be the unnormalized Hodge Laplacian formed using the same weighted inner products. The normalized operator $\widehat{\Delta}_{k,n} = \hat{\delta}_n \hat{d}_n + \bar{\sigma}_{\delta,k}^{-1} \hat{d}_n \hat{\delta}_n$ only rescales the down-Laplacian term. Indeed, if $U = \hat{\delta}_n \hat{d}_n$ and $D = \hat{d}_n \hat{\delta}_n$, then $UD = DU = 0$, and the finite-dimensional Hodge decomposition splits C^k into harmonic, exact, and coexact summands. On these summands, U and D act separately. Hence $\Delta_{k,n}^{\mathrm{std}}$ and $\widehat{\Delta}_{k,n}$ admit a common orthonormal eigenbasis: the normalization rescales the exact eigenvalues by $\bar{\sigma}_{\delta,k}^{-1}$, leaves the coexact and harmonic eigenvalues unchanged.

For $c \in C^k(X_{\epsilon_n}^{(n)})$, we define the *discrete Hodge energy* and the corresponding *discrete Rayleigh quotient* by

$$\widehat{Q}_{k,n}(c) := \|\hat{d}_n c\|_n^2 + \bar{\sigma}_{\delta,k}^{-1} \|\hat{\delta}_n c\|_n^2 \quad \text{and} \quad \widehat{R}_{k,n}(c) := \frac{\widehat{Q}_{k,n}(c)}{\|c\|_n^2} \quad \text{for} \quad c \neq 0.$$

The second term in $\widehat{Q}_{k,n}$ is omitted for $k = 0$. Since $C^k(X_{\epsilon_n}^{(n)})$ is finite dimensional, the usual matrix spectral theorem gives

$$\widehat{\lambda}_{j,n}^{(k)} = \min_{\substack{W \subset C^k(X_{\epsilon_n}^{(n)}) \\ \dim W = j+1}} \max_{0 \neq c \in W} \widehat{R}_{k,n}(c). \tag{5.4}$$

5.2. Spectral Upper Bounds.

The first step is uniform convergence of Rayleigh quotients on each fixed finite-dimensional space of smooth forms.

Lemma 5.3. *Let $V \subset \Omega^k(M)$ be finite dimensional. Then, with probability tending to one, $q_n|_V$ is injective. Moreover,*

$$\sup_{0 \neq \alpha \in V} \left| \widehat{R}_{k,n}(q\alpha) - R_k(\alpha) \right| \xrightarrow{\mathbb{P}} 0. \tag{5.5}$$

Proof. Because V is finite dimensional, all smooth norms are bounded on the unit sphere $\{\alpha \in V : \|\alpha\|_M = 1\}$. The norm convergence

$$\sup_{\substack{\alpha \in V \\ \|\alpha\|_M = 1}} \left| \|q\alpha\|_n^2 - 1 \right| \xrightarrow{\mathbb{P}} 0$$

follows directly from Theorem 4.1 and the normalization of the degree k inner product fixed above. On the event that this supremum is less than $1/2$, every $\alpha \in V$ with $\|\alpha\|_M = 1$ satisfies $\|q\alpha\|_n^2 \geq 1/2$. Hence $q_n|_V$ is injective on an event whose probability tends to one.

For the exterior derivative term in the Hodge energy Q, Lemma 3.1 gives

$$\hat{d}_n(q\alpha) = q(d\alpha).$$

Applying Theorem 4.1 in degree $k+1$, uniformly for $d\alpha$ with α in the unit sphere of V, gives

$$\|\hat{d}_n(q\alpha)\|_n^2 \xrightarrow{\mathbb{P}} \|d\alpha\|_M^2$$

uniformly on that sphere. For the codifferential term, when $m \geq k \geq 1$, applying the normalized pairing convergence fixed above from Theorem 4.5, we get

$$\bar{\sigma}_{\delta,k}^{-1} \|\hat{\delta}_n(q\alpha)\|_n^2 \xrightarrow{\mathbb{P}} \|\delta\alpha\|_M^2$$

uniformly on the unit sphere of V. Combining the two energy terms gives

$$\sup_{\substack{\alpha \in V \\ \|\alpha\|_M = 1}} \left| \widehat{Q}_{k,n}(q\alpha) - Q_k(\alpha) \right| \xrightarrow{\mathbb{P}} 0.$$

Finally, the norm convergence above implies that $\|q\alpha\|_n^2$ is bounded away from zero uniformly on the unit sphere with probability tending to one; dividing the convergent energies by the convergent denominators proves (5.5). □

Theorem 5.4. *For every fixed k and j, we have $\limsup_{n\to\infty} \widehat{\lambda}_{j,n}^{(k)} \leq \lambda_j^{(k)}$ in probability; in particular,*

$$\mathbb{P}\left[\widehat{\lambda}_{j,n}^{(k)} \leq \lambda_j^{(k)} + \eta\right] \to 1 \qquad \textit{for every } \eta > 0.$$

Proof. Let $V_j \subset \Omega^k(M)$ be the span of the first $j+1$ smooth eigenforms of Δ_k, counted with multiplicity. By (5.2),

$$\max_{0 \neq \alpha \in V_j} R_k(\alpha) = \lambda_j^{(k)}.$$

By Lemma 5.3, with probability tending to one,

$$\max_{0 \neq \alpha \in V_j} \widehat{R}_{k,n}(q\alpha) \leq \lambda_j^{(k)} + \eta.$$

By the injectivity of $q_n|_{V_j}$ in Lemma 5.3, we also get that $\dim(q(V_j)) = j+1$ with probability tending to one. On the intersection of these two events, the discrete min-max formula (5.4) gives

$$\widehat{\lambda}_{j,n}^{(k)} \leq \max_{0 \neq c \in q(V_j)} \widehat{R}_{k,n}(c) = \max_{0 \neq \alpha \in V_j} \widehat{R}_{k,n}(q\alpha) \leq \lambda_j^{(k)} + \eta.$$

This proves the claim. □

5.3. Harmonic 1-Forms. In the preceding section, we considered upper bounds for the eigenvalues of the discrete Hodge Laplacian. In order to obtain more precise spectral relationships between continuous and discrete Hodge Laplacians, we require interpolation methods to translate simplicial cochains back to differential forms. As this is beyond the scope of the present article, we focus our attention on 1-Laplacians, where we can leverage known spectral convergence results about graph Laplacians to prove convergence of harmonic 1-forms.

Throughout this section, we assume that M is connected, and our main tool is the spectral convergence of the graph Laplacian (the $k=0$ Laplacian) to the Laplace-Beltrami operator in [9]. Let θ be as in Definition 4.4. Define $\hat{L}_n : C^0(X_{\epsilon_n}) \to C^0(X_{\epsilon_n})$ to be the unnormalized weighted graph Laplacian

$$(\hat{L}_n a)(x_i) := \frac{1}{n\epsilon_n^{m+2}} \sum_{j=1}^n \theta\left(\frac{\|x_i - x_j\|^2}{\epsilon_n^2}\right)(a(x_i) - a(x_j)), \quad a \in C^0(X_{\epsilon_n}),$$

as defined in [9, Equation 2.3]. The degree 0 Laplacian $\hat{\Delta}_{0,n}$ is defined in terms of our inner product in (3.3), which sums over oriented edges. By direct computation,

$$\widehat{\Delta}_{0,n} = \frac{2n}{(n-1)\rho_0\sigma_1^\theta} \hat{L}_n \quad \text{on } C^0(X_{\epsilon_n}). \tag{5.6}$$

Thus the two graph Laplacians have the same kernel, and their Rayleigh quotients differ by a scalar, which is bounded above and below uniformly in n.

Remark 5.5. The graph Laplacian results of [9] use a closed ϵ-graph, with an edge whenever $\|x_i - x_j\| \le \epsilon$, while our Vietoris–Rips convention uses the open condition $\|x_i - x_j\| < \epsilon$. Thus, [9] uses a radial profile $\eta : [0,\infty) \to [0,\infty)$ which is Lipschitz on $[0,1]$ and may be positive at the endpoint. To apply their theorem, we define the radial profile by $\eta(r) = \theta(r^2)$ whenever $r \neq 1$ and set $\eta(1) = \lim_{r\uparrow 1}\theta(r)$. For any $\epsilon > 0$, the event $\|x_i - x_j\| = \epsilon$ has probability zero for each pair (i,j), and hence the closed graph Laplacian built from η_θ agrees almost surely with $\hat{L}_n$. Thus the spectral convergence theorem applies to $\hat{L}_n$ through this endpoint-modified profile.

Proposition 5.6. *Suppose M is connected and ϵ_n satisfies (5.1). For the eigenvalue $\widehat{\lambda}^{(0)}_{1,n}$ from (5.3), there is a constant $c_0 > 0$ such that*

$$\mathbb{P}\left[\widehat{\lambda}^{(0)}_{1,n} \ge c_0\right] \to 1.$$

Proof. Let $\mu_{1,n}$ denote the first positive eigenvalue of $\hat{L}_n$ with respect to the degree-zero cochain inner product. We note that the ϵ_n bandwidth in Theorem 4.5 satisfies the hypotheses of [9, Theorem 2.4]. Thus, by [9, Theorem 2.4], (using the modified profile from Remark 5.5), $\mu_{1,n}$ converges in probability to the first positive eigenvalue of the corresponding continuum analogue of $\hat{L}_n$, multiplied by a constant $c > 0$ determined by θ. Because M is connected, the limiting eigenvalue is positive, $\mu_1 = c\lambda^{(0)}_1 > 0$ for some constant $c > 0$. Hence

$$\mathbb{P}\left[\mu_{1,n} \ge \frac{\mu_1}{2}\right] \to 1.$$

The comparison between Laplacians in (5.6) above gives the corresponding identity for the eigenvalues, and thus with probability tending to one,

$$\widehat{\lambda}^{(0)}_{1,n} = \frac{2n}{(n-1)\rho_0\sigma_1^\theta}\mu_{1,n} \ge \frac{nc}{(n-1)\rho_0\sigma_1^\theta}\lambda^{(0)}_1.$$

Taking any $c_0 < (c\lambda^{(0)}_1)/(\rho_0\sigma_1^\theta)$ proves the claim for all sufficiently large n. □

Corollary 5.7. *Let $u_n \in C^0(X_{\epsilon_n})$ satisfy $u_n \perp \ker \hat{d}_n$ with respect to the degree-zero inner product. With probability tending to one,*

$$\|u_n\|_n^2 \lesssim \|\hat{d}_n u_n\|_n^2.$$

Consequently, if $\|\widehat{\Delta}_{0,n} u_n\|_n \xrightarrow{\mathbb{P}} 0$, then $\|\hat{d}_n u_n\|_n \xrightarrow{\mathbb{P}} 0$.

Proof. On the event in Proposition 5.6, the restriction of $\widehat{\Delta}_{0,n}$ to $(\ker \hat{d}_n)^\perp$ has spectrum contained in $[c_0, \infty)$. Therefore, we get

$$c_0\|u_n\|_n^2 \le \langle \widehat{\Delta}_{0,n}u_n, u_n\rangle_n = \widehat{Q}_{0,n}(u_n) = \|\hat{d}_n u_n\|_n^2.$$

On the same event, we apply Cauchy-Schwarz to get

$$c_0\|u_n\|_n^2 \le \langle \widehat{\Delta}_{0,n}u_n, u_n\rangle_n \le \|\widehat{\Delta}_{0,n}u_n\|_n\|u_n\|_n \quad \text{so that} \quad \|u_n\|_n \le c_0^{-1}\|\widehat{\Delta}_{0,n}u_n\|_n.$$

Using the identity defining the degree-zero energy,

$$\|\hat{d}_n u_n\|_n^2 = \langle \widehat{\Delta}_{0,n}u_n, u_n\rangle_n \le \|\widehat{\Delta}_{0,n}u_n\|_n\|u_n\|_n.$$

The convergence assumption on $\widehat{\Delta}_{0,n}u_n$ gives the desired result. □

Proposition 5.8. *Assume that M is connected. Let $\omega \in \Omega^1(M)$ be a smooth harmonic one-form. For each n, decompose $q\omega$ by the finite-dimensional Hodge decomposition of $C^1(X_{\epsilon_n})$,*

$$q\omega = \hat{\omega}_n + \hat{d}_n \widehat{\beta}_n, \tag{5.7}$$

where $\hat{\omega}_n \in \ker \widehat{\Delta}_{1,n}$ and $\widehat{\beta}_n \perp \ker \hat{d}_n$. Then

$$\|q\omega - \hat{\omega}_n\|_n \xrightarrow{\mathbb{P}} 0.$$

Proof. Since ω is a harmonic 1-form, $d\omega = 0$ and $\delta\omega = 0$. By the cochain property Lemma 3.1, we have $\hat{d}_n(q\omega) = q(d\omega) = 0$. Since the finite-dimensional Hodge decomposition restricts to $\ker \widehat{\Delta}_{1,n} \oplus \operatorname{im} \hat{d}_n$ on $\ker \hat{d}_n$, the cochain $q\omega$ has the decomposition in (5.7) and satisfies $\hat{\delta}_n \hat{\omega}_n = 0$. Next, we apply the codifferential pairing convergence Theorem 4.5 with $\alpha = \beta = \omega$. Since $\delta\omega = 0$, it gives

$$\|\hat{\delta}_n(q\omega)\|_n \xrightarrow{\mathbb{P}} 0,$$

by the codifferential pairing convergence. Applying $\hat{\delta}_n$ to (5.7) and using $\hat{\delta}_n \hat{\omega}_n = 0$, we obtain $\hat{\delta}_n \hat{d}_n \widehat{\beta}_n = \hat{\delta}_n(q\omega)$, so that $\|\widehat{\Delta}_{0,n}\widehat{\beta}_n\|_n \to 0$ in probability. Since $\widehat{\beta}_n \perp \ker \hat{d}_n$, Corollary 5.7 gives

$$\|\hat{d}_n \widehat{\beta}_n\|_n \xrightarrow{\mathbb{P}} 0.$$

The result follows from (5.7). □

5.4. Circular Coordinates. We conclude this section by considering an application of our results to the convergence of the circular coordinates construction of [12]. This construction has a topological step using persistent cohomology, which selects an integral cohomology class on the Vietoris-Rips complex of the point cloud, and an analytic smoothing step, which replaces a chosen integer cocycle by its discrete harmonic representative. The result below concerns the smoothing step, assuming that the topological step has selected the integral class corresponding to the smooth harmonic form.

In our setting, we consider a point cloud X sampled from an embedded compact connected manifold $M \subset \mathbb{R}^N$. Consider a harmonic 1-form $\omega \in \Omega^1(M)$ with integral periods. After choosing a basepoint $p \in M$, this defines the smooth circular coordinate

$$f_\omega : M \to S^1, \qquad f_\omega(y) := \exp\left(2\pi i \int_\gamma \omega\right), \tag{5.8}$$

where γ is any path from the basepoint p to y. The integral-period condition makes this independent of the path. On the event $\mathcal{T}_n$ that X_{ϵ_n} is homotopy equivalent to M (see Proposition 4.8), the class of the discretization $q\omega$ is represented by an integer-valued cocycle $\hat{\alpha}_n$. Because $q\omega - \hat{\alpha}_n$ is exact, there exists $\zeta_n \in C^0(X_{\epsilon_n})$ such that

$$q\omega = \hat{\alpha}_n + \hat{d}_n \zeta_n \quad \text{where} \quad f_\omega(y) = \exp\left(2\pi i \zeta_n(y) + \phi\right) \tag{5.9}$$

for $y \in X$ and some additive constant ϕ with $e^\phi \in S^1$. The integrality of $\hat{\alpha}_n$ is essential here: edge-path sums of $\hat{\alpha}_n$ are integer-valued and therefore disappear after exponentiation. In practice, we do not have access to $q\omega$; instead, we use the selected cocycle $\hat{\alpha}_n$ and $\langle \cdot, \cdot \rangle_n$ to compute the Hodge decomposition and define a discrete approximation to F_ω by

$$\hat{\omega}_n = \hat{\alpha}_n + \hat{d}_n \hat{\zeta}_n \quad \text{and} \quad \hat{f}_n(y) = \exp\left(2\pi i \hat{\zeta}_n(y)\right),$$

where $\hat{\omega}_n$ is the discrete harmonic representative of the real cohomology class of $\hat{\alpha}_n$, equivalently of $q\omega$. The additive phase constant in (5.9) records the choice of basepoint in (5.8). The following result shows that the approximation $\hat{f}_n$ converges in L^2 to f_ω up to a global phase difference with high probability.

Corollary 5.9. *On the event $\mathcal{T}_n$ from Proposition 4.8, suppose that an integer-valued cocycle $\hat{\alpha}_n$ has been selected whose image in real cohomology is the class of $q\omega$. Let*

$$\hat{\omega}_n = \hat{\alpha}_n + \hat{d}_n \hat{\zeta}_n \tag{5.10}$$

be the discrete harmonic representative of this class, and set $\hat{f}_n(x_i) = \exp(2\pi i \hat{\zeta}_n(x_i))$. Then the real lifts satisfy $\inf_{c\in\mathbb{R}} \|\zeta_n - \hat{\zeta}_n - c\|_n \xrightarrow{\mathbb{P}} 0$, and consequently

$$\inf_{\varphi\in\mathbb{R}} \left(\frac{1}{n} \sum_{i=1}^{n} \left| f_\omega(x_i) - e^{2\pi i\varphi} \hat{f}_n(x_i) \right|^2 \right)^{1/2} \xrightarrow{\mathbb{P}} 0. \tag{5.11}$$

Proof. Work on the event in the statement. Since $\hat{\alpha}_n$ and $q\omega$ determine the same real cohomology class, $\hat{\omega}_n$ is also the harmonic term in the Hodge decomposition of $q\omega$. We also restrict to the high-probability analytic events underlying Proposition 5.8 and Corollary 5.7. Subtracting (5.10) from (5.9) gives

$$q\omega - \hat{\omega}_n = \hat{d}_n(\zeta_n - \hat{\zeta}_n).$$

By Proposition 5.8, the left hand side tends to zero in the degree-one norm. Since the topology event makes $X^{(n)}_{\epsilon_n}$ connected, $\ker \hat{d}_n$ consists of constants. Adding a constant to $\hat{\zeta}_n$ preserves (5.10) and only rotates $\hat{f}_n$, so we may assume $\zeta_n - \hat{\zeta}_n \perp \ker \hat{d}_n$. Then, Corollary 5.7 gives

$$\|\zeta_n - \hat{\zeta}_n\|_n \xrightarrow{\mathbb{P}} 0.$$

This proves the asserted convergence of the lifts modulo constants. The vertex norm is equivalent to the empirical $n^{-1}\sum_i$ norm up to fixed constants. Since $t \mapsto e^{2\pi i t}$ is Lipschitz on $\mathbb{R}/\mathbb{Z}$, this implies

$$\left(\frac{1}{n} \sum_{i=1}^{n} \left| e^{2\pi i \zeta_n(x_i)} - e^{2\pi i \hat{\zeta}_n(x_i)} \right|^2 \right)^{1/2} \xrightarrow{\mathbb{P}} 0.$$

By (5.9), exponentiating ζ_n recovers f_ω on the sample vertices up to one phase. The additive constant used to align the lifts only rotates $\hat{f}_n$, which is exactly the infimum over global rotations in (5.11). The objects in the statement are defined on events whose probabilities tend to one; extending $\hat{f}_n$ arbitrarily to the complements does not change convergence in probability. ☐

Remark 5.10. In the case where $\dim H^1(M;\mathbb{R}) = 1$, there is only a single harmonic 1-form ω up to normalization. Thus, on the event $\mathcal{T}_n$ from Proposition 4.8, we get $H^1(X_{\epsilon_n}) = H^1(M)$, and we obtain the class required for Corollary 5.9 by computing the kernel of the discrete Laplacian.

6. Proof of Uniform Convergence of Inner Products

In this section, we will prove Theorem 4.1 by considering a uniform bound over the unit ball $\mathcal{B}_\mathbb{R}$ as defined in (4.1). In particular, we will separately consider the bias and variance terms given in (4.2). In Section 6.1, we estimate the bias using a local normal-coordinate expansion of the simplex integrals and the orthogonal invariance of the admissible kernel. For the variance term, the aim is to apply a concentration bound for U-statistics to obtain a probabilistic bound for fixed forms α and β, and extend this to a uniform bound using a covering argument. In order to obtain improved rates for the covering argument, we begin by replacing the Euclidean U-kernel with a geometric U-kernel in Section 6.2. We consider the pointwise concentration estimate of this geometric kernel in Section 6.3, followed by the uniform estimate in Section 6.4. We complete the proof in Section 6.5. In this section, we fix a smooth symmetric weight function $w : M^{k+1} \to \mathbb{R}$ and an admissible VR kernel κ_ϵ with defining function Θ, and suppress this from the inner product notation.

6.1. Bias Estimate. We begin with the bias term, where we prove the following estimate.

Proposition 6.1. *There exists a constant $\sigma_k^\Theta > 0$ such that for $0 < \epsilon < \epsilon_0$, we have*

$$\sup_{\alpha,\beta \in \mathcal{B}_\mathbb{R}} \left| \mathbb{E}\langle q\alpha, q\beta\rangle^{\kappa,w}_{n,\epsilon} - \sigma_k^\Theta \langle \alpha, \beta\rangle^{P_\Delta}_M \right| \lesssim \epsilon^2,$$

where $P(\boldsymbol{y}) = w(\boldsymbol{y})\rho(y_0)\cdots\rho(y_k) : M^{k+1} \to \mathbb{R}$ and $P_\Delta(x) = P(x,\ldots,x) : M \to \mathbb{R}$.

Here, we consider the expectation of the empirical inner product

$$\mathbb{E}_{\mu^n}[\langle q\alpha, q\beta\rangle_{n,\epsilon}] = \mathbb{E}_{\mu^{k+1}}[h_\epsilon(\alpha,\beta)] = \frac{(k!)^2}{\epsilon^{k(m+2)}} \int_{D^k_\epsilon(M)} q\alpha(\boldsymbol{y}) q\beta(\boldsymbol{y}) \kappa_\epsilon(\boldsymbol{y}) P(\boldsymbol{y})\, d\mathrm{vol}^{k+1}(\boldsymbol{y}),$$

where $h_\epsilon(\alpha,\beta)$ is the U-kernel defined in (3.5), $D^k_\epsilon(M)$ is the fat diagonal defined in (3.6). Note that we can decompose this domain by

$$D^k_\epsilon(M) = \bigcup_{x\in M} \{(x,\boldsymbol{y}) : \boldsymbol{y} \in D^k_\epsilon(M,x)\} \text{ where } D^k_\epsilon(M,x) := \{\boldsymbol{y} \in M^k : \mathrm{diam}_\mathbb{R}(x,\boldsymbol{y}) < \epsilon\},$$

so that the expectation can be expressed as

$$\mathbb{E}[\langle q\alpha, q\beta\rangle^\kappa_{n,\epsilon}] = \int_M \left(\int_{D^k_\epsilon(M,x)} \frac{(k!)^2}{\epsilon^{k(m+2)}} q\alpha(x,\boldsymbol{y}) q\beta(x,\boldsymbol{y}) \kappa_\epsilon(x,\boldsymbol{y}) P(x,\boldsymbol{y}) d\mathrm{vol}^k(\boldsymbol{y}) \right) d\mathrm{vol}(x).$$

Our aim is to bound the error between the pointwise $\langle \alpha_x, \beta_x\rangle_x$ and the inner integral

$$I(\alpha,\beta,x) := \int_{D^k_\epsilon(M,x)} \frac{(k!)^2}{\epsilon^{k(m+2)}} q\alpha(x,\boldsymbol{y}) q\beta(x,\boldsymbol{y}) \kappa_\epsilon(x,\boldsymbol{y}) P(x,\boldsymbol{y}) d\mathrm{vol}^k(\boldsymbol{y}).$$

Since $\epsilon < \epsilon_0$, we apply Lemma 3.3 to approximate the discretizations $q\alpha(x,\boldsymbol{y})$ with evaluations $\alpha_x(\boldsymbol{v})$ where $\boldsymbol{v} = \log^k_x(\boldsymbol{y}) \in T^k_x M$. Under this change of coordinates, the domain becomes

$$L^k_\epsilon(M,x) = \log^k_x(D^k_\epsilon(M,x)) = \{\boldsymbol{v} = (v_1,\ldots,v_k) \in T^k_x M : \mathrm{diam}_\mathbb{R}(x, \exp_x(\boldsymbol{v})) < \epsilon\}.$$

We perform a second change of variables by rescaling $\boldsymbol{u} = \epsilon^{-1}\boldsymbol{v}$, where the rescaled domain is

$$\widetilde{L}^k_\epsilon(M,x) = L^k_\epsilon(M,x)/\epsilon = \{\boldsymbol{u} = (u_1,\ldots,u_k) \in T^k_x M : \mathrm{diam}_\mathbb{R}(x,\exp_x(\epsilon\boldsymbol{u})) < \epsilon\}.$$

We will now need several lemmas to complete the result. First, we record a combined kernel and domain replacement estimate. This is the point where the possible discontinuity of Θ at the boundary of its support is used. We first need to replace the scaled log domain $\tilde{L}^k_\epsilon(M)$ with one that exhibits $O(V)$-invariance (Figure 1).

Lemma 6.2. *Let $\kappa_\epsilon : (\mathbb{R}^N)^{k+1} \to \mathbb{R}$ be an admissible VR kernel of degree k. For $\boldsymbol{u} \in T^k_x M$, set $\tilde{p}(\boldsymbol{u}) = p(0,\boldsymbol{u})$ and*

$$\mathcal{V}^k(M,x) = \{\boldsymbol{u} \in T^k_x M : \mathrm{diam}_{T_xM}(0,\boldsymbol{u}) < 1\}.$$

If $f : T^k_x M \to \mathbb{R}$ is bounded on $\widetilde{L}^k_\epsilon(M,x) \cup \mathcal{V}^k(M,x)$, then, for every $0 < \epsilon < \epsilon_0$,

$$\left| \int_{T^k_x M} f(\boldsymbol{u}) \left(\kappa_\epsilon(x, \exp_x(\epsilon\boldsymbol{u})) \mathbb{1}_{\widetilde{L}^k_\epsilon(M,x)}(\boldsymbol{u}) - \Theta(\tilde{p}(\boldsymbol{u})) \mathbb{1}_{\mathcal{V}^k(M,x)}(\boldsymbol{u}) \right) d\boldsymbol{u} \right| \lesssim \|f\|_{L^\infty(\widetilde{L}^k_\epsilon(M,x)\cup\mathcal{V}^k(M,x))} \epsilon^2, \tag{6.1}$$

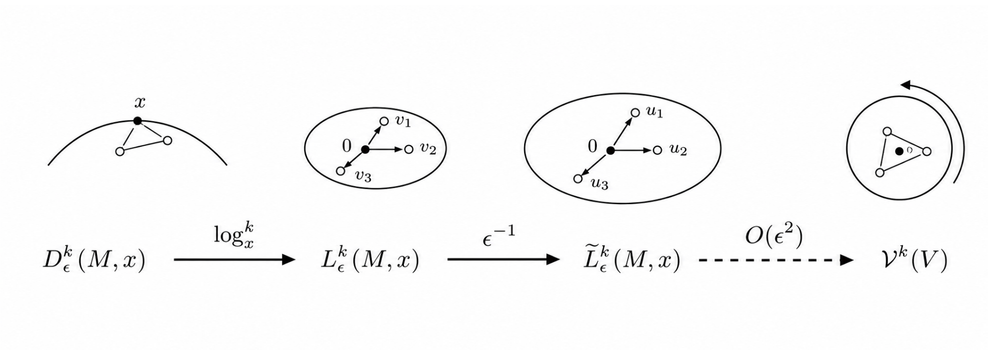

Figure 1. Transformations of the inner-integral region of integration.

uniformly in $x \in M$.

Proof. Set $u_0 = 0$. Both $\widetilde{L}^k_\epsilon(M,x)$ and $\mathcal{V}^k(M,x)$ are contained in a uniformly bounded subset $B \subset T^k_x M$. On this bounded set, the Taylor expansion (2.6) gives, uniformly in x, ϵ, and $\boldsymbol{u}$,

$$\exp_x(\epsilon u_i) - \exp_x(\epsilon u_j) = \epsilon(u_i - u_j) + \epsilon^2(Q_x(u_i,u_i) - Q_x(u_j,u_j)) + O(\epsilon^3).$$

Since $u_i - u_j \in T_x M$ and $Q_x(u_i,u_i) - Q_x(u_j,u_j) \in N_x M$, the quadratic cross term vanishes. Hence

$$\epsilon^{-2}\|\exp_x(\epsilon u_i) - \exp_x(\epsilon u_j)\|^2 = \|u_i - u_j\|^2 + O(\epsilon^2), \qquad 0 \le i < j \le k. \tag{6.2}$$

Consider the set of simplices in the tangent space possessing an edge of squared distance that is order ϵ^2 close to one:

$$S_\epsilon = \bigcup_{i<j} \left\{ \boldsymbol{u} \in B : \left| \|u_i - u_j\|^2 - 1 \right| \lesssim \epsilon^2 \right\}.$$

This is a finite union of bounded domains of thickness $O(\epsilon^2)$, so $\mathrm{vol}(S_\epsilon) \lesssim \epsilon^2$. On $B \setminus S_\epsilon$, the comparison (6.2) implies that membership in $\widetilde{L}^k_\epsilon(M,x)$ and membership in $\mathcal{V}^k(M,x)$ agree. Moreover, either both arguments of Θ lie in the interior of the cube $[0,1)^{\binom{k+1}{2}}$, or both terms in (6.1) vanish. Therefore, using the Lipschitz bound for Θ on the interior of the cube, we get

$$\left| \kappa_\epsilon(x, \exp_x(\epsilon \boldsymbol{u})) \mathbb{1}_{\widetilde{L}^k_\epsilon(M,x)}(\boldsymbol{u}) - \Theta(\tilde{p}(\boldsymbol{u})) \mathbb{1}_{\mathcal{V}^k(M,x)}(\boldsymbol{u}) \right| \lesssim \epsilon^2$$

on $B \setminus S_\epsilon$. On S_ϵ, the integrand can be nonzero only on $S_\epsilon \cap (\widetilde{L}^k_\epsilon(M,x) \cup \mathcal{V}^k(M,x))$; there both terms are uniformly bounded, and the shell has volume $O(\epsilon^2)$. Integrating these two estimates proves the result. $\square$

Lemma 6.3. *Let $w : M^{k+1} \to \mathbb{R}$ be a smooth function. Then for any $x \in M$ and $\boldsymbol{v} = (v_1, \ldots, v_k) \in T^k_x M$ such that $\|\boldsymbol{v}\| < \mathrm{inj}_{\mathbb{R}}(M)$,*

$$w(x, \exp_x(\boldsymbol{v})) - w_\Delta(x) = L(x,\boldsymbol{v}) + R(x,\boldsymbol{v})$$

where $w_\Delta(x) = w(x,\ldots,x)$. Here, $L(x,\boldsymbol{v})$ is linear in $\boldsymbol{v}$ and satisfies $|L(x,\boldsymbol{v})| \le C_1\|\boldsymbol{v}\|$, while $|R(x,\boldsymbol{v})| \le C_2\|\boldsymbol{v}\|^2$, where $C_1, C_2 > 0$ depends on w.

Proof. Let $\tilde{w}_x : T^k_x M \to \mathbb{R}$ be defined by $\tilde{w}_x(\boldsymbol{v}) = w(x, \exp_x(\boldsymbol{v}))$. Note that this is a smooth function on a vector space, so by a Taylor expansion about $\boldsymbol{v} = 0$, we obtain

$$\tilde{w}_x(\boldsymbol{v}) = \tilde{w}_x(0) + d\tilde{w}_x(0)[\boldsymbol{v}] + R(x,\boldsymbol{v}),$$

where $|R_x(\boldsymbol{v})| \leq C_2\|\boldsymbol{v}\|^2$ is a uniform bound over M since M is compact. Furthermore, we let

$$L(x, \boldsymbol{v}) = d\tilde{w}_x(0)[\boldsymbol{v}],$$

which is a linear function of $\boldsymbol{v}$, and thus satisfies $|L(x, \boldsymbol{v})| \leq C_1\|\boldsymbol{v}\|$. □

The next preliminary lemma shows that the integral of $\alpha_x(\boldsymbol{v})\beta_x(\boldsymbol{v})$ against an $O(T_xM)$-invariant kernel over T_x^kM yields a constant multiple of $\langle \alpha_x, \beta_x\rangle_x$. We state a slightly more general form of this result, which will be reused in a later section.

Lemma 6.4. *Let V be an m-dimensional Hilbert space, $H \subset V^\ell$ be a linear subspace invariant under the diagonal $O(V)$-action, and $d\boldsymbol{v}$ be the induced Euclidean measure on H. Let $f : H \to \mathbb{R}$ be invariant under this action, and let $\Phi : H \to \Lambda^r V$ be an $O(V)$-equivariant map such that $|\Phi(\boldsymbol{v})|^2|f(\boldsymbol{v})|$ is integrable over H. Then there exists a constant $\sigma_{\Phi,f}$, depending only on Φ and f, such that for all $\alpha, \beta \in \Lambda^r V^*$,*

$$\int_H \alpha(\Phi(\boldsymbol{v}))\beta(\Phi(\boldsymbol{v}))f(\boldsymbol{v})\, d\boldsymbol{v} = \sigma_{\Phi,f}\langle \alpha, \beta\rangle.$$

More explicitly, if $e^1, \ldots, e^m$ is an orthonormal basis of V^*, then

$$\sigma_{\Phi,f} = \int_H \left[(e^1 \wedge \cdots \wedge e^r)(\Phi(\boldsymbol{v}))\right]^2 f(\boldsymbol{v})\, d\boldsymbol{v}.$$

Proof. Let e^I be a basis for $\Lambda^r V^*$, so that $\alpha = \sum_I \alpha_I e^I$ and $\beta = \sum_J \beta_J e^J$. Let

$$A_{I,J} = \int_H e^I(\Phi(\boldsymbol{v}))e^J(\Phi(\boldsymbol{v}))f(\boldsymbol{v})d\boldsymbol{v}.$$

Our aim is to show that $A_{I,J} = 0$ whenever $I \neq J$ and $A_{I,I} = A_{J,J}$ for all I, J. First, suppose $I \neq J$ so there exists some index $a \in [m]$ such that $a \in I$ but $a \notin J$. Define the reflection $R_a \in O(V)$ such that $R_a(e_a) = -e_a$ and $R_a(e_s) = e_s$ for all $s \neq a$. By changing variables with R_a, and using the invariance of H, the invariance of f, and the equivariance of Φ, we get

$$A_{I,J} = \int_H e^I(R_a\Phi(\boldsymbol{v}))e^J(R_a\Phi(\boldsymbol{v}))f(\boldsymbol{v})d\boldsymbol{v} = -\int_H e^I(\Phi(\boldsymbol{v}))e^J(\Phi(\boldsymbol{v}))f(\boldsymbol{v})d\boldsymbol{v} = -A_{I,J},$$

so that $A_{I,J} = 0$. Next, let $I = (i_1 < \ldots < i_r)$ and $J = (j_1 < \ldots < j_r)$. Choose $U \in O(V)$ such that $U(e_{j_q}) = e_{i_q}$ for all $q \in [r]$, so that $e^I(U\Phi(\boldsymbol{v})) = e^J(\Phi(\boldsymbol{v}))$. Then, by the same invariance and equivariance argument, we get

$$A_{I,I} = \int_H e^I(U\Phi(\boldsymbol{v}))^2 f(\boldsymbol{v})d\boldsymbol{v} = \int_H e^J(\Phi(\boldsymbol{v}))^2 f(\boldsymbol{v})d\boldsymbol{v} = A_{J,J},$$

and we let σ_f denote this value. Then, we obtain

$$\int_H \alpha(\Phi(\boldsymbol{v}))\beta(\Phi(\boldsymbol{v}))f(\boldsymbol{v})d\boldsymbol{v} = \sigma_f \sum_I \alpha_I \beta_I = \sigma_f\langle \alpha, \beta\rangle.$$

□

Next, we can put these lemmas together to obtain the pointwise error estimate.

Lemma 6.5. *For $0 < \epsilon < \epsilon_0$, we have*

$$\sup_{\alpha,\beta \in \mathcal{B}_{\mathbb{R}}} \sup_{x \in M} \left| I(\alpha, \beta, x) - \sigma_k^\Theta \langle \alpha_x, \beta_x\rangle_x P_\Delta(x)\right| \lesssim \epsilon^2.$$

Proof. We begin with a change of coordinates by $\boldsymbol{v} = \log_x(\boldsymbol{y})$ with the Jacobian $\overline{J}(x, \boldsymbol{v}) = \prod_{i=1}^{k} J(x, v_i)$ to get

$$I(\alpha, \beta, x) = \int_{L_\epsilon^k(M,x)} \frac{(k!)^2}{\epsilon^{k(m+2)}} q\alpha(x, \boldsymbol{y}) q\beta(x, \boldsymbol{y}) \kappa_\epsilon(x, \boldsymbol{y}) P(x, \boldsymbol{y}) \overline{J}(x, \boldsymbol{v}) d\boldsymbol{v},$$

where $L_\epsilon^k(M, x) = \log^k(D_\epsilon^k(M, x))$. We further rescale by setting $\boldsymbol{v} = \epsilon \boldsymbol{u}$. We apply Lemma 3.3, Lemma 6.3, and the Jacobian remainder estimate (2.5) to obtain the following uniform bounds (over $\mathcal{B}_{\mathbb{R}}$ and M) on the bounded rescaled domain $\widetilde{L}_\epsilon^k(M, x) = L_\epsilon^k(M, x)/\epsilon$,

$$\begin{aligned} k! q\alpha(x, \boldsymbol{y}) &= \epsilon^k \alpha_x(\boldsymbol{u}) + \epsilon^{k+1} L_\alpha(\boldsymbol{u}) + O(\epsilon^{k+2}) \\ k! q\beta(x, \boldsymbol{y}) &= \epsilon^k \beta_x(\boldsymbol{u}) + \epsilon^{k+1} L_\beta(\boldsymbol{u}) + O(\epsilon^{k+2}) \\ P(x, \boldsymbol{y}) &= P_\Delta(x) + \epsilon L_P(\boldsymbol{u}) + O(\epsilon^2) \\ \overline{J}(x, \boldsymbol{y}) &= 1 + O(\epsilon^2). \end{aligned}$$

We collect the terms of order 1 and ϵ and integrate the rest over $\widetilde{L}_\epsilon^k(M, x)$ to get

$$I(\alpha, \beta, x) = \int_{\widetilde{L}_\epsilon^k(M,x)} \Big(\alpha_x(\boldsymbol{u}) \beta_x(\boldsymbol{u}) P_\Delta(x) + \epsilon \mathcal{L}(\boldsymbol{u}) \Big) \kappa_\epsilon(x, \exp_x(\epsilon \boldsymbol{u})) d\boldsymbol{u} + O(\epsilon^2).$$

Here, $\mathcal{L}(\boldsymbol{u})$ is given by

$$\mathcal{L}(\boldsymbol{u}) = L_\alpha(\boldsymbol{u}) \beta_x(\boldsymbol{u}) P_\Delta(x) + \alpha_x(\boldsymbol{u}) L_\beta(\boldsymbol{u}) P_\Delta(x) + \alpha_x(\boldsymbol{u}) \beta_x(\boldsymbol{u}) L_P(\boldsymbol{u}),$$

which is homogeneous of order $2k+1$, so that $\mathcal{L}(-\boldsymbol{u}) = -\mathcal{L}(\boldsymbol{u})$. Applying Lemma 6.2 with $f(\boldsymbol{u}) = \alpha_x(\boldsymbol{u}) \beta_x(\boldsymbol{u}) P_\Delta(x) + \epsilon \mathcal{L}(\boldsymbol{u})$ gives

$$I(\alpha, \beta, x) = \int_{\mathcal{V}^k(M,x)} \Big(\alpha_x(\boldsymbol{u}) \beta_x(\boldsymbol{u}) P_\Delta(x) \Theta(\tilde{p}(\boldsymbol{u})) + \epsilon \mathcal{L}(\boldsymbol{u}) \Theta(\tilde{p}(\boldsymbol{u})) \Big) d\boldsymbol{u} + O(\epsilon^2).$$

Since $\mathcal{L}(\boldsymbol{u})$ is odd and Θ is even, we get

$$\int_{\mathcal{V}^k(M,x)} \mathcal{L}(\boldsymbol{u}) \Theta(\tilde{p}(\boldsymbol{u})) d\boldsymbol{u} = 0.$$

Then, by applying Lemma 6.4 (in this case $H = V^k$ and $\Phi(\boldsymbol{v}) = v_1 \wedge \ldots \wedge v_k$), using the $O(T_x M)$-invariance of $\Theta(\tilde{p}(\boldsymbol{u})) \mathbb{1}_{\mathcal{V}^k(M,x)}(\boldsymbol{u})$, we get

$$\int_{\mathcal{V}^k(M,x)} \alpha_x(\boldsymbol{u}) \beta_x(\boldsymbol{u}) \Theta(\tilde{p}(\boldsymbol{u})) d\boldsymbol{u} = \sigma_k^\Theta \langle \alpha_x, \beta_x \rangle_x \ \text{ where } \ \sigma_k^\Theta = \int_{\mathcal{V}^k(M,x)} (e^1 \wedge \ldots \wedge e^k)(\boldsymbol{u})^2 \, \Theta(\tilde{p}(\boldsymbol{u})) d\boldsymbol{u},$$

and $\sigma_k^\Theta > 0$ by the positive-measure nonzero hypothesis on Θ in Theorem 4.1, since the squared determinant above is nonzero away from the rank-deficient locus. Therefore,

$$I(\alpha, \beta, x) = \sigma_k^\Theta \langle \alpha_x, \beta_x \rangle_x P_\Delta(x) + O(\epsilon^2).$$

□

6.2. A Geometric U-Kernel and Covering Bound. Next, we consider the variance term in (4.2). For fixed $\alpha, \beta \in \Omega^k(\mathbb{R}^N)$, the empirical inner product can be written as the U-statistic $\langle q\alpha, q\beta \rangle_{n,\epsilon} = \mathbb{U}_n[h_\epsilon(\alpha, \beta)]$ of a U-kernel h_ϵ. To prepare the later uniform concentration argument, we replace the Euclidean U-kernel $h_\epsilon^e = h_\epsilon$ by a geometric U-kernel h_ϵ^g coming from the normal-coordinate expansion of the simplex integrals. The point is that h_ϵ^g differs from h_ϵ^e only by a controlled higher-order error, while depending on the forms only through their restricted ambient 1-jets along M. This reduces the covering argument to an intrinsic covering bound on M rather than $\mathbb{R}^N$.

Let $\boldsymbol{y} = (y_0, \ldots, y_k) \in D^k_\epsilon(M)$. For each $\ell \in \{0, \ldots, k\}$, we define

$$\boldsymbol{v}^\ell = \Big(\log_{y_\ell}(y_0), \ldots, \log_{y_\ell}(y_{\ell-1}), \log_{y_\ell}(y_{\ell+1}), \ldots, \log_{y_\ell}(y_k) \Big) \in T^k_{y_\ell} M.$$

For the Euclidean U-kernel h^e_ϵ, we use Lemma 3.3, where the orientation signs cancel, to obtain for any $\ell \in \{0, \ldots, k\}$,

$$(k!)^2 q\alpha(\boldsymbol{y}) q\beta(\boldsymbol{y}) = \Big(\alpha_{y_\ell}(\boldsymbol{v}^\ell)\beta_{y_\ell}(\boldsymbol{v}^\ell) + \alpha_{y_\ell}(\boldsymbol{v}^\ell) L_{\beta,\ell}(\boldsymbol{v}^\ell) + \beta_{y_\ell}(\boldsymbol{v}^\ell) L_{\alpha,\ell}(\boldsymbol{v}^\ell) \Big) + R(\boldsymbol{y}), \tag{6.3}$$

where $R(\boldsymbol{y}) \lesssim \epsilon^{2k+2}$ uniformly over $\mathcal{B}_\mathbb{R}$. Now, for each $\alpha, \beta \in \Omega^k(\mathbb{R}^N)$, we define a *geometric* U-kernel $h^g_\epsilon(\alpha, \beta) : M^{k+1} \to \mathbb{R}$ by symmetrizing the first order term in (6.3)

$$h^g_\epsilon(\alpha,\beta)(\boldsymbol{y}) := \frac{\epsilon^{-k(m+2)}}{k+1} \sum_{\ell=0}^{k} \Big(\alpha_{y_\ell}(\boldsymbol{v}^\ell)\beta_{y_\ell}(\boldsymbol{v}^\ell) + \alpha_{y_\ell}(\boldsymbol{v}^\ell) L_{\beta,\ell}(\boldsymbol{v}^\ell) + \beta_{y_\ell}(\boldsymbol{v}^\ell) L_{\alpha,\ell}(\boldsymbol{v}^\ell) \Big) \kappa_\epsilon(\boldsymbol{y}) w(\boldsymbol{y}), \tag{6.4}$$

over the choice of basepoint y_ℓ. This has the property that

$$\sup_{\alpha,\beta \in \mathcal{B}_\mathbb{R}} |h^e_\epsilon(\alpha,\beta)(\boldsymbol{y}) - h^g_\epsilon(\alpha,\beta)(\boldsymbol{y})| \lesssim \epsilon^{-k(m+2)} R(\boldsymbol{y}) \mathbb{1}_{D^k_\epsilon(M)}(\boldsymbol{y}) \lesssim \epsilon^{2-km} \mathbb{1}_{D^k_\epsilon(M)}(\boldsymbol{y}), \tag{6.5}$$

where $\mathbb{1}_{D^k_\epsilon(M)}$ denotes the indicator function. The key property of $h^g_\epsilon(\alpha, \beta)$ is that it only depends on the following restricted 1-jet.

Definition 6.6. For an ambient form $\alpha = \sum_I \alpha_I dz^I$, we define the *ambient 1-jet of α on M* to be

$$j(\alpha) = (\alpha_I|_M, \partial_i \alpha_I|_M)_{i,I} \in \mathrm{Lip}(M, E),$$

where $E = \Lambda^k \mathbb{R}^N \oplus (\Lambda^k \mathbb{R}^N)^N$ contains information about the ambient form along with all ambient derivatives, restricted to M. Thus $j : \Omega^k_c(\mathbb{R}^N) \to \mathrm{Lip}(M, E)$.

The following covering bound can be obtained via a standard sup-norm entropy bound for bounded Lipschitz functions on compact m-dimensional domains.

Lemma 6.7. *Let $\mathcal{N}_\gamma$ be the smallest number of $C^0(M, E)$ balls of radius $\gamma > 0$ with centers lying in $j(\mathcal{B}_\mathbb{R})$ required to cover $j(\mathcal{B}_\mathbb{R}) \subset \mathrm{Lip}(M, E)$. Then,*

$$\log \mathcal{N}_\gamma \lesssim \gamma^{-m}.$$

Proof. For $L > 0$, let $\mathrm{Lip}_L(M, E)$ denote the set of maps $F : M \to E$ satisfying

$$\|F\|_{C^0(M,E)} \leq L \quad \text{and} \quad \mathrm{Lip}(F) \leq L.$$

Since $\alpha \in \mathcal{B}_\mathbb{R}$ has uniformly bounded coefficients, first derivatives, and second derivatives, there exists $L > 0$ such that $j(\mathcal{B}_\mathbb{R}) \subset \mathrm{Lip}_L(M, E)$. By the standard sup-norm entropy bound for bounded Lipschitz functions on compact m-dimensional domains, see [40, Example 5.10], there exists a $C^0(M, E)$ cover of $\mathrm{Lip}_L(M, E)$ by $\mathcal{N}^{\mathrm{Lip}}_\gamma$ balls of radius $\gamma/2$, such that $\log(\mathcal{N}^{\mathrm{Lip}}_\gamma) \lesssim \gamma^{-m}$. We retain only those balls which intersect $j(\mathcal{B}_\mathbb{R})$ and denote this subcollection by $\{B'_r\}_{r=1}^{\mathcal{N}'_\gamma}$. For each B'_r, choose $\eta^r \in \mathcal{B}_\mathbb{R}$ such that $j(\eta^r) \in B'_r$, and let B_r be the $C^0(M, E)$ ball of radius γ centered at $j(\eta^r)$. Because each B'_r has radius $\gamma/2$, the balls $\{B_r\}_{r=1}^{\mathcal{N}'_\gamma}$ cover $j(\mathcal{B}_\mathbb{R})$ and have centers in $j(\mathcal{B}_\mathbb{R})$. Therefore,

$$\log \mathcal{N}_\gamma \leq \log \mathcal{N}'_\gamma \leq \log \mathcal{N}^{\mathrm{Lip}}_\gamma \lesssim \gamma^{-m}.$$

$\square$

6.3. Pointwise Variance Estimate. In order to perform the pointwise concentration, our main tool will be Bernstein inequalities for U-statistics [19, 3]. We cite the statement from [34].

Theorem 6.8. [19, 3] *Given a U-kernel h, there exist constants $c > 0$ depending only on k such that*

$$\mathbb{P}\left[|\mathbb{U}_n[h] - \mathbb{E}[h]| > t\right] \leq 4\exp\left(\frac{-cnt^2}{\Xi + \|h\|_\infty t}\right),$$

where $\Xi = \mathrm{Var}_{x\sim\mu}\left[\mathbb{E}_{\boldsymbol{y}\sim\mu^k}[h(x,\boldsymbol{y})]\right]$.

We begin with an immediate lemma which will be used later.

Lemma 6.9. *Let $\mathbb{1}_{D^k_\epsilon(M)} : M^{k+1} \to \mathbb{R}$ be the indicator function on $D^k_\epsilon(M)$, which is symmetric. For a sufficiently large constant C, define*

$$\mathcal{E}_\epsilon := \left\{\mathbb{U}_n[\mathbb{1}_{D^k_\epsilon(M)}] \leq C\epsilon^{km}\right\},$$

Then,

$$\mathbb{P}[\mathcal{E}^c_\epsilon] \leq 4\exp\left(-cn\epsilon^{km}\right).$$

Proof. As $\mathbb{1}_{D^k_\epsilon(M)}$ is an indicator function, we have $\|\mathbb{1}_{D^k_\epsilon(M)}\|_\infty \leq 1$. Furthermore, since $D^k_\epsilon(M,x) \subset (B_\epsilon(x,\mathbb{R}^N)\cap M)^k$, estimates on the size of the intersection balls $B_\epsilon(x,\mathbb{R}^N)\cap M$ yield

$$\mathbb{E}[\mathbb{1}_{D^k_\epsilon(M)}] = \int_{D^k_\epsilon(M)} d\mu^{k+1}(\boldsymbol{y}) \lesssim \epsilon^{km} \quad \text{and} \quad g(x) = \mathbb{E}_{\boldsymbol{y}}[\mathbb{1}_{D^k_\epsilon(M)}(x,\boldsymbol{y})] = \int_{D^k_\epsilon(M,x)} d\mu^k(\boldsymbol{y}) \lesssim \epsilon^{km}.$$

Therefore, we have $\Xi = \mathrm{Var}(g(x)) \lesssim \epsilon^{2km}$. Choosing C large enough so that $\mathbb{E}[\mathbb{1}_{D^k_\epsilon(M)}] + \epsilon^{km} \leq C\epsilon^{km}$, applying Theorem 6.8, and setting $t = \epsilon^{km}$, we get (when $\epsilon < 1$) that

$$\mathbb{P}[\mathcal{E}^c_\epsilon] \leq 4\exp\left(\frac{-cn\epsilon^{km}}{1+\epsilon^{km}}\right) \leq 4\exp\left(-cn\epsilon^{km}\right).$$

□

Now we can use this Bernstein inequality to obtain our form-wise estimate.

Lemma 6.10. *For $\alpha, \beta \in \mathcal{B}_\mathbb{R}$ we have*

$$\mathbb{P}\left[\left|\mathbb{U}_n[h^g_\epsilon(\alpha,\beta)] - \mathbb{E}[h^g_\epsilon(\alpha,\beta)]\right| > t\right] \leq 4\exp\left(\frac{-cnt^2}{1+\epsilon^{-km}t}\right)$$

Proof. We begin by bounding $h^g_\epsilon(\alpha,\beta)$ (6.4) itself. The support of $h^g_\epsilon(\alpha,\beta)(\boldsymbol{y})$ is $D^k_\epsilon(M)$, where we have the bounds $|\alpha_x(v)| \lesssim \epsilon^k$ and $|L_\alpha(v)| \lesssim \epsilon^{k+1}$, so that

$$|h^g_\epsilon(\alpha,\beta)(\boldsymbol{y})| \lesssim \epsilon^{-km}\mathbb{1}_{D^k_\epsilon(M)} \quad \text{and} \quad \|h^g_\epsilon(\alpha,\beta)\|_\infty \lesssim \epsilon^{-km}.$$

For the first projection $g(x) = \mathbb{E}_{\boldsymbol{y}\sim\mu^k}[h^g_\epsilon(\alpha,\beta)(x,\boldsymbol{y})]$, the same pointwise bound gives

$$|g(x)| \lesssim \epsilon^{-km}\mu^k(D^k_\epsilon(M,x)) \lesssim 1,$$

since $\mu^k(D^k_\epsilon(M,x)) \lesssim \epsilon^{km}$. Thus, the Bernstein projection variance satisfies $\Xi = \mathrm{Var}(g(x)) \lesssim 1$. Then, we obtain our result by applying Theorem 6.8.

□

6.4. Uniform Variance Estimate. Next, we will prove the covering argument.

Lemma 6.11. *We have*

$$\mathbb{P}\left[\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\mathbb{U}_n[h^g_\epsilon(\alpha,\beta)] - \mathbb{E}[h^g_\epsilon(\alpha,\beta)]\right| \gtrsim t+\gamma\right] \leq 4\mathcal{N}^2\exp\left(\frac{-c_1nt^2}{1+\epsilon^{-km}t}\right) + 4\exp(-c_2n\epsilon^{km}).$$

Proof. Let $\{B_r\}_{r=1}^{\mathcal{N}}$ be a collection of $\mathcal{C}^0(M,E)$-balls $B_r \subset \mathrm{Lip}(M,E)$ of radius γ which cover $j(\mathcal{B}_{\mathbb{R}})$ from Lemma 6.7. Let $\eta^r \in \Omega^k(\mathbb{R}^N)$ be the ambient form where $j(\eta^r)$ is the center of B_r. By applying a union bound to Lemma 6.10, we get

$$\mathbb{P}\left[\sup_{r,s\in[\mathcal{N}]} \left|\mathbb{U}_n[h^g_\epsilon(\eta^r,\eta^s)] - \mathbb{E}[h^g_\epsilon(\eta^r,\eta^s)]\right| > t\right] \leq 4\mathcal{N}^2\exp\left(\frac{-c_1nt^2}{1+\epsilon^{-km}t}\right) \tag{6.6}$$

Let $\alpha,\beta\in\mathcal{B}_{\mathbb{R}}$. Because the B_r cover $j(\mathcal{B}_{\mathbb{R}})$, there exist η^r,η^s such that

$$\|j(\alpha)-j(\eta^r)\|_{\mathcal{C}^0(M,E)} < \gamma \quad \text{and} \quad \|j(\beta)-j(\eta^s)\|_{\mathcal{C}^0(M,E)} < \gamma.$$

Let $\boldsymbol{y} = (y_0,\ldots,y_k)\in M^{k+1}$. We note that for any $\ell\in\{0,\ldots,k\}$, we have

$$|\alpha_{y_\ell}(\boldsymbol{v}^\ell) - \eta^r_{y_\ell}(\boldsymbol{v}^\ell)| \leq \|j(\alpha)-j(\eta^r)\|_{\mathcal{C}^0(M,E)}\|\boldsymbol{v}^\ell\|^k \quad \text{and} \quad |L_\alpha(\boldsymbol{v}^\ell) - L_{\eta^r}(\boldsymbol{v}^\ell)| \leq \|j(\alpha)-j(\eta^r)\|_{\mathcal{C}^0(M,E)}\|\boldsymbol{v}^\ell\|^{k+1},$$

and similarly for β and η^s. Thus, we can bound the difference of the geometric U-kernels by

$$|h^g_\epsilon(\alpha,\beta)(\boldsymbol{y}) - h^g_\epsilon(\eta^r,\eta^s)(\boldsymbol{y})| \lesssim \frac{\gamma}{\epsilon^{km}}\mathbb{1}_{D^k_\epsilon(M)}, \tag{6.7}$$

and because $\mu^{k+1}(D^k_\epsilon(M)) \lesssim \epsilon^{km}$, we get

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\mathbb{E}[h^g_\epsilon(\alpha,\beta) - h^g_\epsilon(\eta^r,\eta^s)]\right| \lesssim \gamma$$

where the choice of η^r and η^s are dependent on α,β. Applying Lemma 6.9 to (6.7) gives

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\mathbb{U}_n[h^g_\epsilon(\alpha,\beta) - h^g_\epsilon(\eta^r,\eta^s)]\right| \lesssim \frac{\gamma}{\epsilon^{km}}\mathbb{U}_n[\mathbb{1}_{D^k_\epsilon(M)}] \lesssim \gamma.$$

on the event $\mathcal{E}_\epsilon$. Then by the triangle inequality, on the event $\mathcal{E}_\epsilon$,

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\mathbb{U}_n[h^g_\epsilon(\alpha,\beta)] - \mathbb{E}[h^g_\epsilon(\alpha,\beta)]\right| \leq \sup_{r,s\in[\mathcal{N}]} \left|\mathbb{U}_n[h^g_\epsilon(\eta^r,\eta^s)] - \mathbb{E}[h^g_\epsilon(\eta^r,\eta^s)]\right| + C_1\gamma$$

Finally, by applying (6.6) and the probability of $\mathbb{P}[\mathcal{E}^c_\epsilon]$ from Lemma 6.9, we get

$$\mathbb{P}\left[\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\mathbb{U}_n[h^g_\epsilon(\alpha,\beta)] - \mathbb{E}[h^g_\epsilon(\alpha,\beta)]\right| > t+C_1\gamma\right] \leq 4\mathcal{N}^2\exp\left(\frac{-c_1nt^2}{1+\epsilon^{-km}t}\right) + 4\exp(-c_2n\epsilon^{km}).$$

□

Our variance results thusfar have been in terms of geometric kernels, which contain intrinsic information unavailable to the data science practitioner. We apply Lemma 6.7 to obtain the main variance bound for the Euclidean U-kernels as follows.

Proposition 6.12. *Let $p\in(0,1)$. Suppose $\log(8/p)\leq n\epsilon^{km}$. With probability at least $1-p$, we have*

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} |\mathbb{U}_n[h_\epsilon(\alpha,\beta)] - \mathbb{E}[h_\epsilon(\alpha,\beta)]| \lesssim \sqrt{\frac{\gamma^{-m}+\log(\frac{1}{p})}{n}} + \frac{\gamma^{-m}+\log(\frac{1}{p})}{n\epsilon^{km}} + \gamma + \epsilon^2$$

PROOF. We begin by bounding the error between the Euclidean and geometric U-kernels. By (6.5), we note that

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} |\mathbb{E}[h^e_\epsilon(\alpha,\beta) - h^g_\epsilon(\alpha,\beta)]| \lesssim \epsilon^2 \quad \text{and} \quad \sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} |\mathbb{U}_n[h^e_\epsilon(\alpha,\beta) - h^g_\epsilon(\alpha,\beta)]| \lesssim \epsilon^2,$$

where the latter holds on the event $\mathcal{E}_\epsilon$ defined in Lemma 6.9. We note that this is exactly the support event used in Lemma 6.11, so its failure probability is charged only once. Using that lemma, along with the triangle inequality, we obtain

$$\mathbb{P}\left[\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} |\mathbb{U}_n[h_\epsilon(\alpha,\beta)] - \mathbb{E}[h_\epsilon(\alpha,\beta)]| \gtrsim t + \gamma + \epsilon^2\right] \le 4\mathcal{N}^2 \exp\left(\frac{-c_1 n t^2}{1+\epsilon^{-km} t}\right) + 4\exp(-c_2 n\epsilon^{km}).$$

We consider the two bounds on the probability separately. For the second term, if $\log(1/p) \lesssim n\epsilon^{km}$, then we have $4\exp(-c_2 n\epsilon^{km}) \le p/2$. For the first term, we use the covering bound in Lemma 6.7 to get $\log(\mathcal{N}_\gamma) \lesssim \gamma^{-m}$. Then, we use the standard Bernstein inversion and set

$$t \gtrsim \sqrt{\frac{\gamma^{-m} + \log(\frac{1}{p})}{n}} + \frac{\gamma^{-m} + \log(\frac{1}{p})}{n\epsilon^{km}} \quad \text{to get} \quad 4\mathcal{N}^2 \exp\left(\frac{-c_1 n t^2}{1+\epsilon^{-km} t}\right) \le p/2.$$

Thus, with probability at least $1-p$, we have

$$\sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} |\mathbb{U}_n[h_\epsilon(\alpha,\beta)] - \mathbb{E}[h_\epsilon(\alpha,\beta)]| \lesssim \sqrt{\frac{\gamma^{-m} + \log(\frac{1}{p})}{n}} + \frac{\gamma^{-m} + \log(\frac{1}{p})}{n\epsilon^{km}} + \gamma + \epsilon^2$$

□

6.5. Uniform Convergence Proof. Finally, we can put our bias and variance estimates together, and balance the parameters ϵ, γ and n to prove the main result of this section.

PROOF OF THEOREM 4.1. First, we combine the bias estimate from Proposition 6.1 and the variance estimate from Proposition 6.12. In particular, assuming the conditions of Proposition 6.12, with probability at least $1-p$, we have

$$\xi = \sup_{\alpha,\beta\in\mathcal{B}_{\mathbb{R}}} \left|\langle q\alpha, q\beta\rangle^{\kappa,w}_{n,\epsilon} - \sigma^{\Theta}_k \langle\alpha,\beta\rangle^{P_\Delta}_M\right| \lesssim \sqrt{\frac{\gamma^{-m} + \log(\frac{1}{p})}{n}} + \frac{\gamma^{-m} + \log(\frac{1}{p})}{n\epsilon^{km}} + \gamma + \epsilon^2,$$

where we use the fact that $\langle q\alpha, q\beta\rangle^{\kappa,w}_{n,\epsilon} = \mathbb{U}_n[h_\epsilon(\alpha,\beta)]$. Now, we wish to balance the parameters γ, ϵ and n to obtain the desired rate. Here, we fix some $p \in (0,1)$, and let $\ell = \log(1/p)$, so that

$$\xi \lesssim \sqrt{\frac{\gamma^{-m}}{n}} + \sqrt{\frac{\ell}{n}} + \frac{\gamma^{-m}}{n\epsilon^{km}} + \frac{\ell}{n\epsilon^{km}} + \gamma + \epsilon^2.$$

By balancing the terms which involve γ, we obtain

$$\gamma \asymp \max\left\{n^{-1/(m+2)}, (n\epsilon^{km})^{-1/(m+1)}\right\}.$$

Next, we balance the bias ϵ^2 term with the dominant ϵ-dependent stochastic term $(n\epsilon^{km})^{-1/(m+1)}$, which gives us

$$\epsilon \asymp n^{-1/(km+2m+2)}$$

and thus

$$\gamma \asymp n^{-r_{k,m}} \quad \text{where} \quad r_{k,m} = \min\left\{\frac{1}{m+2}, \frac{2}{km+2m+2}\right\}.$$

Thus, the error is

$$\xi \lesssim n^{-r_{k,m}} + \sqrt{\frac{\log(1/p)}{n}} + \frac{\log(1/p)}{n^{(2m+2)/(km+2m+2)}}.$$

□

7. Proof of Codifferential Convergence

In this section, we will prove Theorem 4.5 by expanding the discrete codifferential, and considering bias and variance estimates, similar to Section 6. Unlike the previous section, we will assume that the sampling distribution μ is uniform throughout. We begin with an explicit formula for the discrete codifferential. The following computation is standard for the codifferential in a weighted simplicial complex; for instance see [20] or [16, Lemma 3.6].

Lemma 7.1. *For $a \in C^k(X_\epsilon)$ and $\boldsymbol{y} \in S_\epsilon^{k-1}(X)$, we have*

$$\hat{\delta}a(\boldsymbol{y}) = \frac{k+1}{n-k}\frac{k^2}{\epsilon^{m+2}} \sum_{\substack{x \in X - \boldsymbol{y} \\ (x,\boldsymbol{y}) \in S_\epsilon^k(X)}} a(x,\boldsymbol{y})\frac{\kappa_{\theta,\epsilon}^k(x,\boldsymbol{y})}{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}.$$

Our aim is to show that the discrete codifferential energy converges to the smooth codifferential energy up to a constant. In particular, we fix a pullback form $\alpha = \pi^*\tilde{\alpha}$ and show that

$$\langle \hat{\delta}q\alpha, \hat{\delta}q\alpha\rangle_{n,\epsilon} \to \rho_0^{k+2}\sigma_{\delta,k}^\theta\langle \delta\tilde{\alpha}, \delta\tilde{\alpha}\rangle_M$$

for some constant $\sigma_{\delta,k}^\theta$. To obtain Theorem 4.5, we will conclude at the end using a polarization identity. We will also consider uniform estimates across the intrinsic unit balls

$$\tilde{\mathcal{B}}_M = \{\tilde{\alpha} \in \Omega^k(M) : \|\tilde{\alpha}\|_{C^2(M)} \leq 1\} \quad \text{and} \quad \mathcal{B}_M = \{\pi^*\tilde{\alpha} : \tilde{\alpha} \in \tilde{\mathcal{B}}_M\}.$$

For a fixed pullback form $\alpha = \pi^*\tilde{\alpha}$, we define

$$A_\epsilon^\alpha(x,\boldsymbol{y}) = \epsilon^{-(m+2)}q\alpha(x,\boldsymbol{y})\frac{\kappa_{\theta,\epsilon}^k(x,\boldsymbol{y})}{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}.$$

Using Lemma 7.1, we can decompose

$$\langle \hat{\delta}q\alpha, \hat{\delta}q\alpha\rangle_{n,\epsilon} = D_n^\alpha + F_n^\alpha$$

where D_n^α and F_n^α denote the *diagonal* and *off-diagonal* sums,

$$D_n^\alpha = \binom{n}{k}^{-1} k^2((k+1)!)^2\frac{\epsilon^{-(k-1)(m+2)}}{(n-k)^2} \sum_{\substack{I=(i_1<\ldots<i_k) \\ p \in [n]-I}} A_\epsilon^\alpha(x_p,\boldsymbol{x}_I)^2\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{x}_I), \tag{7.1}$$

$$F_n^\alpha = \binom{n}{k}^{-1} k^2((k+1)!)^2\frac{\epsilon^{-(k-1)(m+2)}}{(n-k)^2} \sum_{\substack{I=(i_1<\ldots<i_k) \\ p_1 \neq p_2 \in [n]-I}} A_\epsilon^\alpha(x_{p_1},\boldsymbol{x}_I)A_\epsilon^\alpha(x_{p_2},\boldsymbol{x}_I)\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{x}_I). \tag{7.2}$$

We will often omit the superscript α in A_ϵ^α, D_n^α, and F_n^α when the form is fixed. In these expressions, the additional factor of $\kappa_{\theta,\epsilon}^{k-1}$ arises from the degree $k-1$ inner product. The proof is then split into three parts.

(1) In Proposition 7.3, we show that the bias of F_n^α converges to $\rho_0^{k+2}\sigma_{\delta,k}^\theta\langle\delta\alpha,\delta\alpha\rangle_M$.
(2) We show that the diagonal term D_n is of lower order.
(3) We bound the variance $F_n - \mathbb{E}[F_n]$ by applying Bernstein concentration for U-statistics.

The following simplex expansion will be used throughout the codifferential proof.

Lemma 7.2. *For $0 < \epsilon < \epsilon_0$, the following holds. Let $(x, \boldsymbol{y}) \in D_\epsilon^k(M)$, let $v_i = \log_x(y_i)$, and write $\boldsymbol{v} = (v_1, \dots, v_k)$. If $\alpha = \pi^*\tilde{\alpha}$ is a pullback form, then*

$$q\alpha(x, \boldsymbol{y}) = \frac{1}{k!}\alpha_x(\boldsymbol{v}) + \frac{1}{(k+1)!}\sum_{i=1}^{k}(\nabla_{v_i}\alpha)_x(\boldsymbol{v}) + R(\boldsymbol{v}),$$

where $|R(\boldsymbol{v})| \lesssim \|\tilde{\alpha}\|_{C^2(M)}\|\boldsymbol{v}\|^{k+2}$.

Proof. This follows from Lemma 3.3 applied to pullback forms: the second fundamental form term vanishes since $\alpha = \pi^*\tilde{\alpha}$ annihilates normal vectors along M, and tangent ambient derivatives $\bar{\nabla}$ restrict to intrinsic covariant derivatives ∇. ☐

7.1. Off-Diagonal Bias Estimate.

The expectation of the off-diagonal term F_n (7.2) is

$$\mathbb{E}[F_n] = \rho_0^{k+2}\frac{k^2((k+1)!)^2(n-k-1)}{\epsilon^{(k-1)(m+2)}(n-k)}\int_{D_\epsilon^{k-1}(M)} I(\boldsymbol{y})^2 d\mathrm{vol}^k(\boldsymbol{y}), \tag{7.3}$$

where we absorb the additional factor of $\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})$ from (7.2) into the normalized, kernel-weighted average of the simplex integral $q\alpha(x, \boldsymbol{y})$ over all possible cones points x that form an ϵ-simplex with $\boldsymbol{y}$:

$$I(\boldsymbol{y}) = \frac{1}{\epsilon^{m+2}}\int_{D_\epsilon(\boldsymbol{y})} q\alpha(x, \boldsymbol{y})\frac{\kappa_{\theta,\epsilon}^k(x, \boldsymbol{y})}{\sqrt{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}} d\mathrm{vol}(x) \quad \text{and} \quad D_\epsilon(\boldsymbol{y}) = \{x \in M : \mathrm{diam}_{\mathbb{R}}(x, \boldsymbol{y}) < \epsilon\}. \tag{7.4}$$

This subsection proves the following deterministic bias estimate for this off-diagonal term.

Proposition 7.3. *Uniformly for $\alpha \in \mathcal{B}_M$, we have*

$$\mathbb{E}[F_n] = \frac{n-k-1}{n-k}\rho_0^{k+2}k^{(m+4)/2}\sigma_\lambda\langle\delta\alpha, \delta\alpha\rangle_M + O(\epsilon),$$

where $\sigma_\lambda > 0$ depends only on m, k, and θ.

The proof uses the Karcher mean [23] to choose a canonical basepoint for the base simplex $\boldsymbol{y} = (y_1, \dots, y_k)$. The Karcher mean is the Riemannian analogue of the Euclidean centroid, characterized by the fact that the tangent vectors from it to the vertices of $\boldsymbol{y}$ balance to zero. By [14, Lemma 3] and [2, Theorem 4.3.1], we have the following.

Proposition 7.4. *For every $0 < \epsilon < \epsilon_0$ and every $\boldsymbol{y} \in D_\epsilon^{k-1}(M)$, the function*

$$E_{\boldsymbol{y}} : M \to \mathbb{R}, \qquad E_{\boldsymbol{y}}(z) = \frac{1}{2}\sum_{i=1}^{k} d_M(z, y_i)^2,$$

has a unique minimizer $z = z(\boldsymbol{y})$, called the Karcher mean, *in a geodesically convex ball containing $\boldsymbol{y}$. This minimizer depends smoothly on $\boldsymbol{y}$ and satisfies*

$$\sum_{i=1}^{k}\log_{z(\boldsymbol{y})}(y_i) = 0.$$

For the rest of the proof, fix $\boldsymbol{y} \in D_\epsilon^{k-1}(M)$ and let $z = z(\boldsymbol{y})$ be its Karcher mean. We write

$$u_i = \epsilon^{-1} \log_z(y_i), \qquad \boldsymbol{u} = (u_1, \ldots, u_k), \qquad \hat{\boldsymbol{u}}_i = (u_1, \ldots, \hat{u}_i, \ldots, u_k),$$

so that $\sum_i u_i = 0$. Let $H_z \subseteq T_z^k M$ be the subspace of centered k-tuples in $T_z M$, and define

$$H_z = \left\{ \boldsymbol{u} \in T_z^k M : \sum_{i=1}^k u_i = 0 \right\} \quad \text{and} \quad \mathcal{V}(\boldsymbol{u}) = \{ w \in T_z M : s(w, \boldsymbol{u}) := \operatorname{diam}(w, \boldsymbol{u})^2 \leq 1 \}, \tag{7.5}$$

where $s(w, \boldsymbol{u})$ notation is used to denote the squared diameter of the tuple $(w, \boldsymbol{u})$. Our aim will be to replace the kernel defined using ambient distances with a normalized kernel $K_{\boldsymbol{u}} : T_z M \to \mathbb{R}$ based on tangent distances, where $K_{\boldsymbol{u}}(w) := 0$ when $s(\boldsymbol{u}) \geq 1$ and

$$K_{\boldsymbol{u}}(w) := \frac{\theta(s(w, \boldsymbol{u}))}{\sqrt{\theta(s(\boldsymbol{u}))}} \quad \text{when} \quad s(\boldsymbol{u}) < 1.$$

with the cases preventing a vanishing denominator. Our first step is to approximate the $q\alpha(x, \boldsymbol{y})$ term in $I(\boldsymbol{y})$ using the discretizations $q\alpha(z, x, \hat{\boldsymbol{y}}_i)$ on simplices based at the Karcher mean $z = z(\boldsymbol{y})$.

Lemma 7.5. *With the notation above,*

$$I(\boldsymbol{y}) = \epsilon^{-(m+2)} \sum_{i=1}^k (-1)^i I_i(\boldsymbol{y}) + O(\epsilon^k) \quad \textit{where} \quad I_i(\boldsymbol{y}) = \int_{D_\epsilon(\boldsymbol{y})} q\alpha(z, x, \hat{\boldsymbol{y}}_i) \frac{\kappa_{\theta,\epsilon}^k(x, \boldsymbol{y})}{\sqrt{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}} d\mathrm{vol}(x). \tag{7.6}$$

Proof. Consider the Euclidean $(k+1)$-simplex with ordered vertices $(z, x, y_1, \ldots, y_k)$. By Lemma 3.1 we get

$$q(d\alpha)(z, x, \boldsymbol{y}) = \hat{d}(q\alpha)(z, x, \boldsymbol{y}) = q\alpha(x, \boldsymbol{y}) - q\alpha(z, \boldsymbol{y}) + \sum_{i=1}^k (-1)^{i+1} q\alpha(z, x, \hat{\boldsymbol{y}}_i).$$

We first bound the two terms not appearing in (7.6). Applying Lemma 7.2 to $q\alpha(z, \boldsymbol{y})$, the leading term and first-order term vanish because $\sum_i u_i = 0$. Hence $|q\alpha(z, \boldsymbol{y})| \lesssim \epsilon^{k+2}$. Similarly, applying the same expansion to the $(k+1)$-form $d\alpha$ shows that $|q(d\alpha)(z, x, \boldsymbol{y})| \lesssim \epsilon^{k+2}$: the tangent vectors $(\epsilon w, \epsilon u_1, \ldots, \epsilon u_k)$ are linearly dependent because $\sum_i u_i = 0$. Therefore,

$$q\alpha(x, \boldsymbol{y}) = \sum_{i=1}^k (-1)^i q\alpha(z, x, \hat{\boldsymbol{y}}_i) + O(\epsilon^{k+2}).$$

Substituting this into (7.4) and using $\mathrm{vol}(D_\epsilon(\boldsymbol{y})) \lesssim \epsilon^m$ gives the result. $\square$

Analogously to Lemma 6.2 in the inner product convergence proof, the next lemma replaces the ambient kernel and integration domain by their tangent-space models away from the base-simplex boundary. Note that $\mathcal{V}(\boldsymbol{u})$ from (7.5) is support of $K_{\boldsymbol{u}}$.

Lemma 7.6. *Let*

$$\widetilde{L}_\epsilon(\boldsymbol{y}) = \{ w \in T_z M : \operatorname{diam}_{\mathbb{R}}(\exp_z(\epsilon w), \boldsymbol{y}) < \epsilon \}.$$

There is a constant $C_0 > 0$ such that, if $s(\boldsymbol{u}) \leq 1 - C_0 \epsilon^2$ and $f : T_z M \to \mathbb{R}$ is bounded on $\mathcal{V}(\boldsymbol{u}) \cup \widetilde{L}_\epsilon(\boldsymbol{y})$, then

$$\left| \int_{T_z M} f(w) \left(\frac{\kappa_{\theta,\epsilon}^k(\exp_z(\epsilon w), \boldsymbol{y})}{\sqrt{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}} \mathbb{1}_{\widetilde{L}_\epsilon(\boldsymbol{y})}(w) - K_{\boldsymbol{u}}(w) \right) dw \right| \lesssim \|f\|_{L^\infty(\mathcal{V}(\boldsymbol{u}) \cup \widetilde{L}_\epsilon(\boldsymbol{y}))} \epsilon^2.$$

Proof. Both $\mathcal{V}(\boldsymbol{u})$ and $\widetilde{L}_\epsilon(\boldsymbol{y})$ are contained in a uniformly bounded subset of T_zM. On this bounded set, by (6.2),

$$\epsilon^{-2}\|\exp_z(\epsilon w) - y_i\|^2 = \|w - u_i\|^2 + O(\epsilon^2).$$

The same comparison applied to pairs of base vertices gives $\epsilon^{-2}s_{k-1}(\boldsymbol{y}) = s(\boldsymbol{u}) + O(\epsilon^2)$. Therefore

$$\left|\epsilon^{-2}s_k(\exp_z(\epsilon w), \boldsymbol{y}) - s(w, \boldsymbol{u})\right| \lesssim \epsilon^2.$$

Since $s(\boldsymbol{u}) \leq 1 - C_0\epsilon^2$ and C_0 is chosen large enough, the denominator $\sqrt{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}$ is compared inside $[0,1)$ to $\sqrt{\theta(s(\boldsymbol{u}))}$, and the Lipschitz bound for θ together with the lower bound c_θ gives an $O(\epsilon^2)$ pointwise error on the common support. The possible disagreement of supports is contained in the shell

$$\left\{w : |s(w, \boldsymbol{u}) - 1| \lesssim \epsilon^2\right\},$$

which has volume $O(\epsilon^2)$ in the uniformly bounded region under consideration. Since θ is bounded, the shell contributes the second $O(\epsilon^2)$ error. $\square$

The following lemma is the local trace mechanism which produces the codifferential. It is analogous to the symmetry argument in Lemma 6.4, but after fixing the centered base tuple $\boldsymbol{u}$, the kernel $K_{\boldsymbol{u}}(w)$ is invariant only under the subgroup of $O(T_zM)$ fixing $W_{\boldsymbol{u}} = \operatorname{span}\{u_1, \ldots, u_k\}$ pointwise. This residual symmetry kills the leading moment and makes the second moment scalar on $W_{\boldsymbol{u}}^\perp$. By Karcher centering, $\sum_i u_i = 0$, so alternation removes the $W_{\boldsymbol{u}}$-directions and leaves the transverse trace in the normal-coordinate formula for $\delta\alpha$.

Lemma 7.7. *For any $i \in \{1, \ldots, k\}$, we have*

$$\int_{T_zM} \alpha_z(w, \hat{\boldsymbol{u}}_i) K_{\boldsymbol{u}}(w) dw = 0 \tag{7.7}$$

and

$$\int_{T_zM} (\nabla_w \alpha)_z(w, \hat{\boldsymbol{u}}_i) K_{\boldsymbol{u}}(w) dw = -\lambda_\theta(\boldsymbol{u})(\delta\alpha)_z(\hat{\boldsymbol{u}}_i),$$

where, for any unit vector $e^\perp \in W_{\boldsymbol{u}}^\perp$ and $W_{\boldsymbol{u}} = \operatorname{span}\{u_1, \ldots, u_k\}$,

$$\lambda_\theta(\boldsymbol{u}) := \int_{T_zM} \langle w, e^\perp\rangle^2 K_{\boldsymbol{u}}(w) dw.$$

Proof. Let $G \subset O(T_zM)$ be the subgroup fixing $W_{\boldsymbol{u}}$ pointwise. Since $K_{\boldsymbol{u}}(w)$ is invariant under the action $K_{\boldsymbol{u}}(w) \mapsto K_{\boldsymbol{u}}(gw)$ for $g \in G$, the vector

$$w_0 = \int_{T_zM} w\, K_{\boldsymbol{u}}(w) dw$$

is fixed by G and hence lies in $W_{\boldsymbol{u}}$. The k vectors $w_0, u_1, \ldots, \hat{u}_i, \ldots, u_k$ all lie in $W_{\boldsymbol{u}}$, which has dimension at most $k-1$, so they are linearly dependent. This proves (7.7) by

$$\int_{T_zM} \alpha_z(w, \hat{\boldsymbol{u}}_i) K_{\boldsymbol{u}}(w) dw = \alpha_z\Big(\int_{T_zM} w K_{\boldsymbol{u}}(w) dw, \hat{\boldsymbol{u}}_i\Big) = \alpha_z(w_0, \hat{\boldsymbol{u}}_i) = 0.$$

For the second identity, let $B(v_1, v_2) = (\nabla_{v_1}\alpha)_z(v_2, \hat{\boldsymbol{u}}_i)$, and let $e_1, \ldots, e_r$ and $e_{r+1}, \ldots, e_m$ be orthonormal bases of $W_{\boldsymbol{u}}$ and $W_{\boldsymbol{u}}^\perp$. Then

$$\int_{T_zM} B(w, w) K_{\boldsymbol{u}}(w) dw = \sum_{p,q} M_{pq} B(e_p, e_q),$$

where

$$M_{pq} = \int_{T_zM} \langle w, e_p\rangle \langle w, e_q\rangle K_{\boldsymbol{u}}(w)dw.$$

If $e_p, e_q \in W_{\boldsymbol{u}}$, then $B(e_p, e_q) = 0$ because $e_q, u_1, \ldots, \hat{u}_i, \ldots, u_k$ are linearly dependent. Mixed terms vanish by reflecting the $W_{\boldsymbol{u}}^\perp$ component. On $W_{\boldsymbol{u}}^\perp$, the same reflection and rotation argument gives $M_{pq} = \lambda_\theta(\boldsymbol{u})\delta_{pq}$. Therefore,

$$\int_{T_zM} (\nabla_w \alpha)_z(w, \hat{\boldsymbol{u}}_i) K_{\boldsymbol{u}}(w)dw = \lambda_\theta(\boldsymbol{u}) \sum_{e_p \in W_{\boldsymbol{u}}^\perp} (\nabla_{e_p}\alpha)_z(e_p, \hat{\boldsymbol{u}}_i).$$

Finally, the normal-coordinate formula for the codifferential [22, Eq. 3.3.47] gives

$$(\delta\alpha)_z(\hat{\boldsymbol{u}}_i) = -\sum_{p=1}^m (\nabla_{e_p}\alpha)_z(e_p, \hat{\boldsymbol{u}}_i) = -\sum_{e_p \in W_{\boldsymbol{u}}^\perp} (\nabla_{e_p}\alpha)_z(e_p, \hat{\boldsymbol{u}}_i),$$

because the summands with $e_p \in W_{\boldsymbol{u}}$ vanish as above. □

We now compute $I(\boldsymbol{y})$ away from the boundary layer $|s(\boldsymbol{u}) - 1| \lesssim \epsilon^2$. Changing variables $x = \exp_z(\epsilon w)$ in $I_i(\boldsymbol{y})$ gives

$$I_i(\boldsymbol{y}) = \epsilon^m \int_{\widetilde{L}_\epsilon(\boldsymbol{y})} q\alpha(z, x, \hat{\boldsymbol{y}}_i) \frac{\kappa^k_{\theta,\epsilon}(x, \boldsymbol{y})}{\sqrt{\kappa^{k-1}_{\theta,\epsilon}(\boldsymbol{y})}} J(z, \epsilon w)dw.$$

Using Lemma 7.6, the Jacobian estimate $J(z, \epsilon w) = 1 + O(\epsilon^2)$, and $|q\alpha(z, x, \hat{\boldsymbol{y}}_i)| \lesssim \epsilon^k$, we obtain

$$I_i(\boldsymbol{y}) = \epsilon^m \int_{T_zM} q\alpha(z, x, \hat{\boldsymbol{y}}_i) K_{\boldsymbol{u}}(w)dw + O(\epsilon^{m+k+2}).$$

By Lemma 7.2,

$$q\alpha(z, x, \hat{\boldsymbol{y}}_i) = \frac{\epsilon^k}{k!}\alpha_z(w, \hat{\boldsymbol{u}}_i) + \frac{\epsilon^{k+1}}{(k+1)!}\left((\nabla_w\alpha)_z(w, \hat{\boldsymbol{u}}_i) + \sum_{j\neq i} (\nabla_{u_j}\alpha)_z(w, \hat{\boldsymbol{u}}_i)\right) + O(\epsilon^{k+2}).$$

The first term vanishes by Lemma 7.7. The third term also vanishes by applying the first identity in Lemma 7.7 to the k-form $(\nabla_{u_j}\alpha)_z$. The second term gives

$$I_i(\boldsymbol{y}) = -\frac{\epsilon^{m+k+1}}{(k+1)!}\lambda_\theta(\boldsymbol{u})(\delta\alpha)_z(\hat{\boldsymbol{u}}_i) + O(\epsilon^{m+k+2}). \tag{7.8}$$

Combining (7.8) with (7.6), we get

$$I(\boldsymbol{y}) = \frac{\epsilon^{k-1}}{(k+1)!}\lambda_\theta(\boldsymbol{u}) \sum_{i=1}^k (-1)^{i+1}(\delta\alpha)_z(\hat{\boldsymbol{u}}_i) + O(\epsilon^k) = \frac{\epsilon^{k-1}}{(k+1)!}\lambda_\theta(\boldsymbol{u})(\delta\alpha)_z(\Phi(\boldsymbol{u})) + O(\epsilon^k), \tag{7.9}$$

where $\Phi(\boldsymbol{u}) = (u_2 - u_1) \wedge \cdots \wedge (u_k - u_1)$, since by direct computation, we have

$$\sum_{i=1}^k (-1)^{i+1}(\delta\alpha)_z(\hat{\boldsymbol{u}}_i) = (\delta\alpha)_z(\Phi(\boldsymbol{u})) = (\delta\alpha)_z(u_2 - u_1, \ldots, u_k - u_1).$$

It remains to integrate the leading term in (7.9) over $D^{k-1}_\epsilon(M)$.

Lemma 7.8. *Uniformly for $\alpha \in \mathcal{B}_M$,*

$$\int_{D^{k-1}_\epsilon(M)} I(\boldsymbol{y})^2 d\mathrm{vol}^k(\boldsymbol{y}) = \frac{k^{m/2}\epsilon^{(k-1)(m+2)}}{((k+1)!)^2} \int_M \int_{H_z} (\delta\alpha)_z(\Phi(\boldsymbol{u}))^2 \lambda_\theta(\boldsymbol{u})^2 d\boldsymbol{u} d\mathrm{vol}(z) + O(\epsilon^{(k-1)(m+2)+1}).$$

Proof. By Proposition 7.4, the change of variables $(z, \boldsymbol{u}) \mapsto \boldsymbol{y}$, with $y_i = \exp_z(\epsilon u_i)$ and $\boldsymbol{u} \in H_z$, is smooth for $0 < \epsilon < \epsilon_0$. Its linearization is $(\xi, \eta) \mapsto (\xi, \dots, \xi) + \epsilon\eta$, and therefore

$$d\mathrm{vol}^k(\boldsymbol{y}) = k^{m/2}\epsilon^{(k-1)m}(1 + O(\epsilon^2))d\boldsymbol{u}d\mathrm{vol}(z).$$

The comparison $\epsilon^{-2}s_{k-1}(\boldsymbol{y}) = s(\boldsymbol{u}) + O(\epsilon^2)$ shows that the image of $D_\epsilon^{k-1}(M)$ is contained in $\{s(\boldsymbol{u}) < 1 + C\epsilon^2\}$ in these coordinates. On the region $s(\boldsymbol{u}) \leq 1 - C_0\epsilon^2$, the expansion (7.9) gives

$$I(\boldsymbol{y})^2 = \frac{\epsilon^{2k-2}}{((k+1)!)^2}\lambda_\theta(\boldsymbol{u})^2(\delta\alpha)_z(\Phi(\boldsymbol{u}))^2 + O(\epsilon^{2k-1}),$$

uniformly for $\alpha \in \mathcal{B}_M$. The boundary layer $\{|s(\boldsymbol{u}) - 1| \lesssim \epsilon^2\}$ has volume $O(\epsilon^2)$ inside the uniformly bounded coordinate region. On this layer the Stokes reduction and the same symmetry argument applied to the exact normalized tangent kernel give the uniform bound $|I(\boldsymbol{y})| \lesssim \epsilon^{k-1}$, so its total contribution is $O(\epsilon^{(k-1)(m+2)+2})$. The Jacobian error contributes the same or a smaller order, proving the lemma. □

Now, we are ready to complete the proof of Proposition 7.3.

Proof of Proposition 7.3. By Lemma 6.4, applied with $H = H_z$ and the $O(T_zM)$-equivariant map $\Phi : H_z \to \Lambda^{k-1}T_zM$, there exists a constant σ_λ such that

$$\int_{H_z} (\delta\alpha)_z(\Phi(\boldsymbol{u}))^2\lambda_\theta(\boldsymbol{u})^2 d\boldsymbol{u} = \sigma_\lambda\langle(\delta\alpha)_z, (\delta\alpha)_z\rangle_z.$$

Moreover, $\sigma_\lambda > 0$ because $\lambda_\theta(\boldsymbol{u}) > 0$ on a positive-measure subset of $\{s(\boldsymbol{u}) < 1\}$, and the trace constant is obtained by applying the preceding identity to a nonzero $(k-1)$-form. Integrating the above identity over the normalized volume form gives

$$\int_M \int_{H_z} (\delta\alpha)_z(\Phi(\boldsymbol{u}))^2\lambda_\theta(\boldsymbol{u})^2 d\boldsymbol{u}d\mathrm{vol}(z) = \sigma_\lambda\langle\delta\alpha, \delta\alpha\rangle_M.$$

Finally, Lemma 7.8 gives

$$\int_{D_\epsilon^{k-1}(M)} I(\boldsymbol{y})^2 d\mathrm{vol}^k(\boldsymbol{y}) = \frac{k^{m/2}\epsilon^{(k-1)(m+2)}}{((k+1)!)^2}\sigma_\lambda\langle\delta\alpha, \delta\alpha\rangle_M + O(\epsilon^{(k-1)(m+2)+1}).$$

Substituting this into (7.3) proves Proposition 7.3:

$$\mathbb{E}[F_n] = \frac{n-k-1}{n-k}\rho_0^{k+2}k^{(m+4)/2}\sigma_\lambda\langle\delta\alpha, \delta\alpha\rangle_M + O(\epsilon).$$

□

7.2. Diagonal Term.

Now, we will consider the diagonal term (7.1).

Lemma 7.9. *On the event $\mathcal{E}_\epsilon$ from Lemma 6.9, we have, for $n \geq 2k$,*

$$\sup_{\alpha \in \mathcal{B}_M} D_n(\alpha) \lesssim \frac{1}{n\epsilon^{m+2}}.$$

Proof. By the elementary bound $|q\alpha(x, \boldsymbol{y})| \lesssim \epsilon^k$ for $\alpha \in \mathcal{B}_M$, the support of θ, and the lower bound $\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y}) \geq c_\theta$ whenever $\kappa_{\theta,\epsilon}^k(x, \boldsymbol{y}) \neq 0$, we have

$$|A_\epsilon^\alpha(x, \boldsymbol{y})| = \left|\epsilon^{-(m+2)}q\alpha(x, \boldsymbol{y})\frac{\kappa_{\theta,\epsilon}^k(x, \boldsymbol{y})}{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}\right| \lesssim \epsilon^{k-m-2}\mathbb{1}_{D_\epsilon^k(M)}(x, \boldsymbol{y}).$$

Therefore,

$$D_n^\alpha \lesssim \binom{n}{k}^{-1} \frac{\epsilon^{-(k-1)(m+2)}}{(n-k)^2} \sum_{\substack{I=(i_1<\cdots<i_k)\\ p\in[n]-I}} \epsilon^{2k-2m-4} \mathbb{1}_{D_\epsilon^k(M)}(x_p, \boldsymbol{x}_I).$$

Each unordered $(k+1)$-tuple is counted $k+1$ times in the last sum, and hence

$$\binom{n}{k}^{-1} \frac{1}{(n-k)^2} \sum_{\substack{I=(i_1<\cdots<i_k)\\ p\in[n]-I}} \mathbb{1}_{D_\epsilon^k(M)}(x_p, \boldsymbol{x}_I) = \frac{1}{n-k}\mathbb{U}_n[\mathbb{1}_{D_\epsilon^k(M)}].$$

On the event $\mathcal{E}_\epsilon$, this gives

$$D_n^\alpha \lesssim \frac{1}{n-k}\epsilon^{-(k-1)(m+2)}\epsilon^{2k-2m-4}\epsilon^{km} \lesssim \frac{1}{n\epsilon^{m+2}},$$

uniformly over $\alpha \in \mathcal{B}_M$. □

7.3. Variance Term. Now, our aim is to bound the variance term $F_n - \mathbb{E}[F_n]$. We do this by first establishing F_n as an order $k+2$ U-statistic. In particular, for $\alpha = \pi^*\tilde{\alpha} \in \Omega^k(U_\epsilon)$, define $h_\epsilon^F(\alpha) : M^{k+2} \to \mathbb{R}$ by

$$h_\epsilon^F(\alpha)(y_1, \ldots, y_{k+2}) = \frac{k^2((k+1)!)^2\epsilon^{-(k-1)(m+2)}}{(k+1)(k+2)} \sum_{i\neq j} A_\epsilon^\alpha(y_i, \hat{\boldsymbol{y}}_{i,j}) A_\epsilon^\alpha(y_j, \hat{\boldsymbol{y}}_{i,j}) \kappa_{\theta,\epsilon}^{k-1}(\hat{\boldsymbol{y}}_{i,j}), \tag{7.10}$$

where $\hat{\boldsymbol{y}}_{i,j} = (y_1, \ldots, \hat{y}_i, \ldots, \hat{y}_j, \ldots, y_{k+2})$. Then,

$$F_n^\alpha = \frac{n-k-1}{n-k}\mathbb{U}_n[h_\epsilon^F(\alpha)]. \tag{7.11}$$

Lemma 7.10. *For $\alpha \in \mathcal{B}_M$, we have*

$$\mathbb{P}[|F_n^\alpha - \mathbb{E}[F_n^\alpha]| > t] \leq 4\exp\left(\frac{-cnt^2}{A\epsilon^{-2} + B\epsilon^{-m(k+1)-2}t}\right),$$

where $A, B > 0$ depend only on M, N, k, and θ.

Proof. First, because $|A_\epsilon^\alpha(x, \boldsymbol{y})| \lesssim \epsilon^{k-m-2}$, we have

$$\|h_\epsilon^F(\alpha)\|_\infty \lesssim \epsilon^{-m(k+1)-2}.$$

Next, we bound the first projection

$$g(x) = \mathbb{E}_{\boldsymbol{y}\sim\mu^{k+1}}[h_\epsilon^F(\alpha)(x, \boldsymbol{y})].$$

From the definition of h_ϵ^F in (7.10), there are two cases to consider. First, suppose x is one of the two distinguished variables. Then, after integrating the other distinguished variable, we use (7.4) to obtain the bound

$$\begin{aligned} &\epsilon^{-(k-1)(m+2)} \int A_\epsilon^\alpha(x, \boldsymbol{y}) A_\epsilon^\alpha(z, \boldsymbol{y}) \kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})\, d\,\mathrm{vol}(z) d\,\mathrm{vol}^k(\boldsymbol{y}) \\ &\lesssim \epsilon^{-(k-1)(m+2)}\epsilon^{k-m-2}\epsilon^{k-1}\epsilon^{km} = \epsilon^{-1}, \end{aligned}$$

where the integration is over the support of the integrand. After integrating in z, (7.4) gives the factor $\sqrt{\kappa_{\theta,\epsilon}^{k-1}(\boldsymbol{y})}I(\boldsymbol{y})$; since the kernel is uniformly bounded, we used $|I(\boldsymbol{y})| \lesssim \epsilon^{k-1}$ and

$\mathrm{vol}(D^k_\epsilon(M,x)) \lesssim \epsilon^{km}$. Second, suppose x is part of the base k-tuple. Then

$$\epsilon^{-(k-1)(m+2)} \int A^\alpha_\epsilon(z_1,x,\boldsymbol{y}) A^\alpha_\epsilon(z_2,x,\boldsymbol{y}) \kappa^{k-1}_{\theta,\epsilon}(x,\boldsymbol{y})\, d\,\mathrm{vol}(z_1) d\,\mathrm{vol}(z_2) d\,\mathrm{vol}^{k-1}(\boldsymbol{y})$$
$$\lesssim \epsilon^{-(k-1)(m+2)} \int_{D^{k-1}_\epsilon(M,x)} I(x,\boldsymbol{y})^2\, d\,\mathrm{vol}^{k-1}(\boldsymbol{y}) \lesssim \epsilon^{-(k-1)(m+2)} \epsilon^{2k-2} \epsilon^{(k-1)m} = 1,$$

Here (7.4), applied to the base k-tuple $(x,\boldsymbol{y})$, identifies the z_1,z_2-integral with $I(x,\boldsymbol{y})^2$, and the same definition together with the lower bound for $\kappa^{k-1}_{\theta,\epsilon}$ on the support of the numerator gives $|I(x,\boldsymbol{y})| \lesssim \epsilon^{k-1}$. Therefore, $\|g\|_\infty \lesssim \epsilon^{-1}$, and

$$\Xi_F = \mathrm{Var}(g(x)) \leq \mathbb{E}[g(x)^2] \lesssim \epsilon^{-2}.$$

Applying Theorem 6.8 to $\mathbb{U}_n[h^F_\epsilon(\alpha)]$, and using (7.11), proves the stated bound after changing constants. □

We will also use a covering bound for the intrinsic unit ball.

Lemma 7.11. *Let $\mathcal{N}_\gamma$ be the smallest number of $\mathcal{C}^0(M)$-balls of radius $\gamma > 0$, with centers in $\tilde{\mathcal{B}}_M$, required to cover $\tilde{\mathcal{B}}_M$. Then*

$$\log \mathcal{N}_\gamma \lesssim \gamma^{-m/2}.$$

Proof. This follows from the classical entropy estimate for Hölder balls [39, Theorem 2.7.1]. Applying this estimate in a finite atlas and finite local trivializations of $\Lambda^k T^* M$ gives the stated bound for the $\mathcal{C}^2(M)$ unit ball. As in Lemma 6.7, after forming the cover we retain only balls meeting $\tilde{\mathcal{B}}_M$ and recenter them at points of $\tilde{\mathcal{B}}_M$. □

Proposition 7.12. *Let $p \in (0,1)$ and suppose $\log(8/p) \leq n\epsilon^{(k+1)m}$. With probability at least $1-p$, we have*

$$\sup_{\alpha \in \mathcal{B}_M} |F^\alpha_n - \mathbb{E}[F^\alpha_n]| \lesssim \sqrt{\frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^2}} + \frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^{m(k+1)+2}} + \gamma\epsilon^{-2}.$$

Proof. Let $\{B_r\}_{r=1}^{\mathcal{N}}$ be a collection of $\mathcal{C}^0(M)$-balls of radius γ which cover $\tilde{\mathcal{B}}_M$, with centers $\tilde{\eta}^r \in \tilde{\mathcal{B}}_M$. Let $\eta^r = \pi^* \tilde{\eta}^r$. By a union bound and Lemma 7.10,

$$\mathbb{P}\left[\sup_{r \in [\mathcal{N}]} |F^{\eta^r}_n - \mathbb{E}[F^{\eta^r}_n]| > t\right] \leq 4\mathcal{N} \exp\left(\frac{-cnt^2}{A\epsilon^{-2} + B\epsilon^{-m(k+1)-2} t}\right). \tag{7.12}$$

Let $\alpha = \pi^* \tilde{\alpha} \in \mathcal{B}_M$. Choose η^r such that $\|\tilde{\alpha} - \tilde{\eta}^r\|_{\mathcal{C}^0(M)} < \gamma$. Then, by the definition of A_ϵ and the lower bound $\kappa^{k-1}_{\theta,\epsilon}(\boldsymbol{y}) \geq c_\theta$ on the support of $\kappa^k_{\theta,\epsilon}(x,\boldsymbol{y})$,

$$|A^\alpha_\epsilon(x,\boldsymbol{y}) - A^{\eta^r}_\epsilon(x,\boldsymbol{y})| \lesssim \gamma\epsilon^{k-m-2} \mathbb{1}_{D^k_\epsilon(M)}(x,\boldsymbol{y}).$$

If $\mathcal{D}_{F,\epsilon}(M)$ denotes the support of h^F_ϵ, then $\mathcal{D}_{F,\epsilon}(M) \subset D^{k+1}_{2\epsilon}(M)$. Hence

$$|h^F_\epsilon(\alpha) - h^F_\epsilon(\eta^r)| \lesssim \gamma\epsilon^{-m(k+1)-2} \mathbb{1}_{D^{k+1}_{2\epsilon}(M)}.$$

Taking expectations gives

$$\sup_{\alpha \in \mathcal{B}_M} |\mathbb{E}[h^F_\epsilon(\alpha) - h^F_\epsilon(\eta^r)]| \lesssim \gamma\epsilon^{-2}.$$

By the same argument as Lemma 6.9, applied to $D^{k+1}_{2\epsilon}(M)$, there is an event

$$\mathcal{F}_\epsilon := \left\{\mathbb{U}_n[\mathbb{1}_{D^{k+1}_{2\epsilon}(M)}] \leq C\epsilon^{(k+1)m}\right\}$$

such that $\mathbb{P}[\mathcal{F}_\epsilon^c] \le 4\exp(-cn\epsilon^{(k+1)m})$. This is distinct from $\mathcal{E}_\epsilon$, which controls the $D_\epsilon^k(M)$ support. On $\mathcal{F}_\epsilon$,

$$\sup_{\alpha\in\mathcal{B}_M} |\mathbb{U}_n[h_\epsilon^F(\alpha) - h_\epsilon^F(\eta^r)]| \lesssim \gamma\epsilon^{-2}.$$

Combining this with (7.12) and the triangle inequality gives

$$\mathbb{P}\left[\sup_{\alpha\in\mathcal{B}_M} |F_n^\alpha - \mathbb{E}[F_n^\alpha]| > t + C\gamma\epsilon^{-2}\right] \le 4\mathcal{N}\exp\left(\frac{-cnt^2}{A\epsilon^{-2} + B\epsilon^{-m(k+1)-2}t}\right) + 4\exp(-cn\epsilon^{(k+1)m}).$$

Using Lemma 7.11 and the standard Bernstein inversion with

$$t \gtrsim \sqrt{\frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^2}} + \frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^{m(k+1)+2}}$$

proves the result. $\square$

We can now finish the proof of the codifferential convergence theorem.

Proof of Theorem 4.5. First, we prove the quadratic estimate uniformly over $\mathcal{B}_M$. Let

$$\xi = \sup_{\alpha\in\mathcal{B}_M} \left|\langle\hat{\delta}q\alpha, \hat{\delta}q\alpha\rangle_{n,\epsilon} - \rho_0^{k+2}\sigma_{\delta,k}^\theta\langle\delta\alpha,\delta\alpha\rangle_M\right|, \qquad \sigma_{\delta,k}^\theta = k^{(m+4)/2}\sigma_\lambda.$$

Combining Proposition 7.3, Lemma 7.9, and Proposition 7.12, and using the concentration event for $\mathbb{U}_n[\mathbb{1}_{D_\epsilon^k(M)}]$ from Lemma 6.9, gives with probability at least $1-p$,

$$\xi \lesssim \epsilon + \frac{1}{n} + \frac{1}{n\epsilon^{m+2}} + \sqrt{\frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^2}} + \frac{\gamma^{-m/2} + \log(1/p)}{n\epsilon^{m(k+1)+2}} + \gamma\epsilon^{-2}, \tag{7.13}$$

provided $\log(12/p) \lesssim n\epsilon^{(k+1)m}$. The factor $\log(12/p)$ comes from allocating failure probability $p/3$ to each prefactor-4 tail: the finite-cover Bernstein tail, the $D_{2\epsilon}^{k+1}(M)$ support tail, and the $D_\epsilon^k(M)$ diagonal support tail.

We now balance γ and ϵ. Put $a = m/2$. The terms involving γ are balanced by taking

$$\gamma \asymp \max\left\{n^{-1/(a+2)}\epsilon^{2/(a+2)}, \, (n\epsilon^{m(k+1)})^{-1/(a+1)}\right\}.$$

With this choice, the dominant ϵ-dependent stochastic term is balanced with the bias term by

$$\epsilon \asymp n^{-1/(m(k+1)+3a+3)} = n^{-2/(2mk+5m+6)}.$$

Thus, for $\epsilon_n \asymp n^{-r_{k,m}^\delta}$ with $r_{k,m}^\delta = 2/(2mk+5m+6)$, the non-logarithmic terms in (7.13) are bounded by $n^{-r_{k,m}^\delta}$, while

$$n\epsilon_n^{(k+1)m} = n^{(3m+6)/(2mk+5m+6)}.$$

Therefore, with probability at least $1-p$,

$$\xi \lesssim n^{-r_{k,m}^\delta} + \frac{\sqrt{\log(1/p)}}{n^{1/2-r_{k,m}^\delta}} + \frac{\log(1/p)}{n^{(3m+2)/(2mk+5m+6)}}.$$

The stated bilinear estimate follows by polarization. Indeed, the error

$$B_n(\alpha,\beta) = \langle\hat{\delta}q\alpha, \hat{\delta}q\beta\rangle_{n,\epsilon_n} - \rho_0^{k+2}\sigma_{\delta,k}^\theta\langle\delta\alpha,\delta\beta\rangle_M$$

is a symmetric bilinear form in α and β, so

$$B_n(\alpha, \beta) = \frac{1}{4}\left(B_n(\alpha+\beta, \alpha+\beta) - B_n(\alpha-\beta, \alpha-\beta)\right).$$

Applying the quadratic estimate to $(\alpha \pm \beta)/\|\alpha \pm \beta\|_{C^2(M)}$ and then rescaling gives the normalized supremum bound in Theorem 4.5. □

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